

**Course: IT3021 – Data Warehousing and Business
Intelligence**

Assignment 02



Student Number – IT23621060

Batch – Kandy UNI

Topic: Cinema Hall Ticket Sales and Customer Behavior

Introduction:

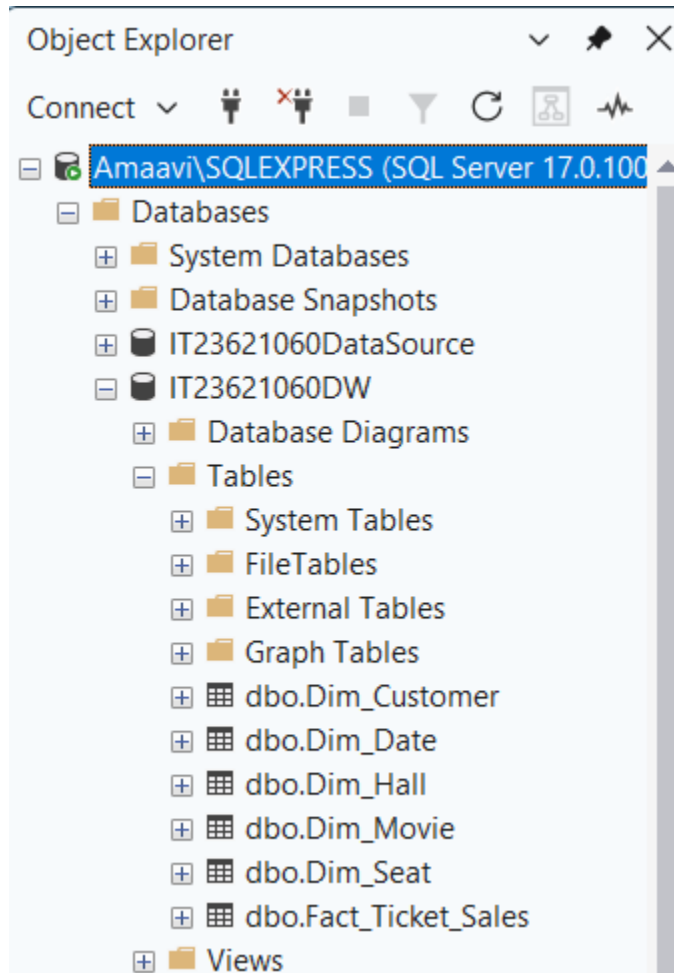
This assignment continues the data warehouse developed in Assignment 1 for the topic Cinema Hall Ticket Sales and Customer Behavior. The existing warehouse was used to implement an SSAS cube, perform OLAP analysis in Excel, and develop interactive reports in Power BI. The final solution supports analytical reporting on cinema ticket sales, halls, movies, dates, and customer-related behavior.

Step 1: Data source for the assignment 2

The data source used for Assignment 2 is the SQL Server data warehouse created in Assignment 1. It follows a star schema design with Fact_Ticket_Sales as the central fact table and Dim_Customer, Dim_Movie, Dim_Hall, Dim_Seat, and Dim_Date as dimensions. The warehouse stores transaction measures such as amount, quantity, and transaction process time, which are used for OLAP analysis and reporting.

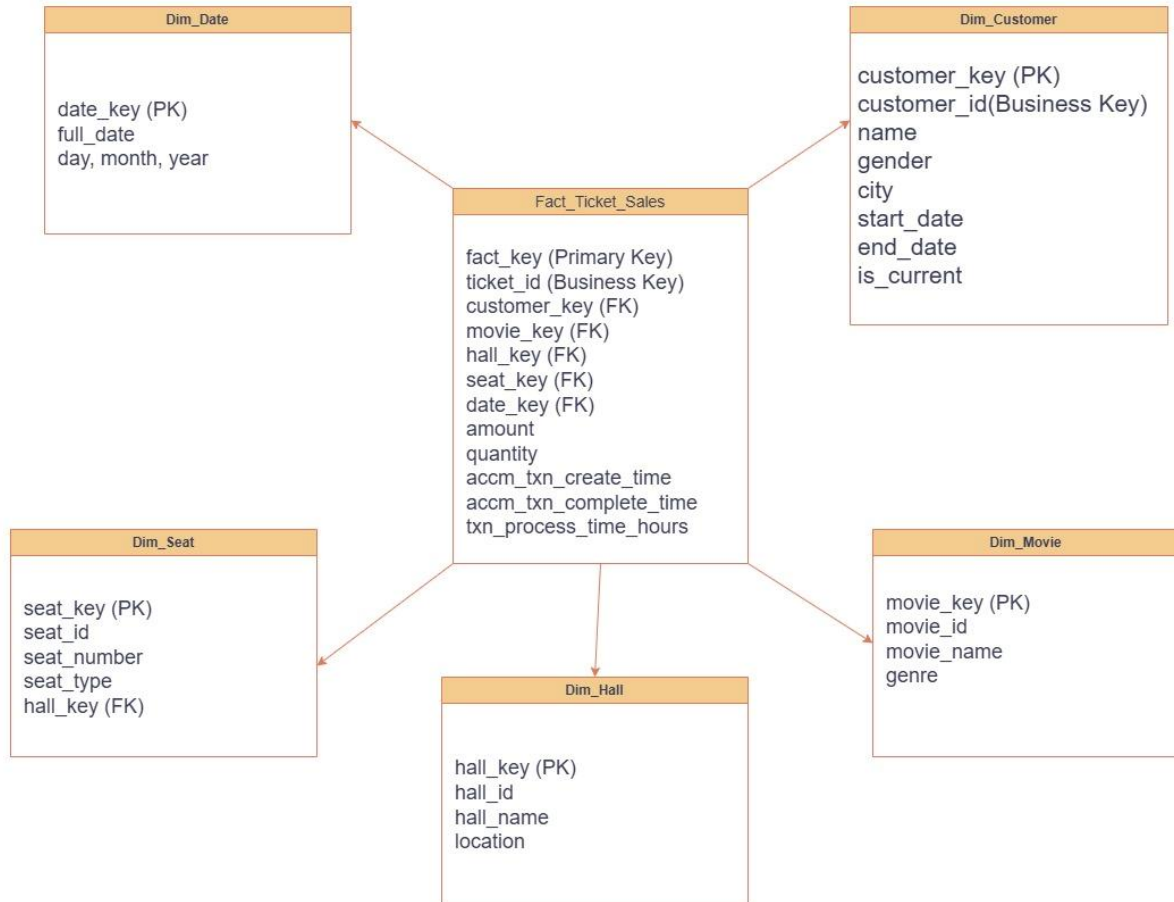
DW database

- Assignment 1 data warehouse used as the source for Assignment 2.



Star schema

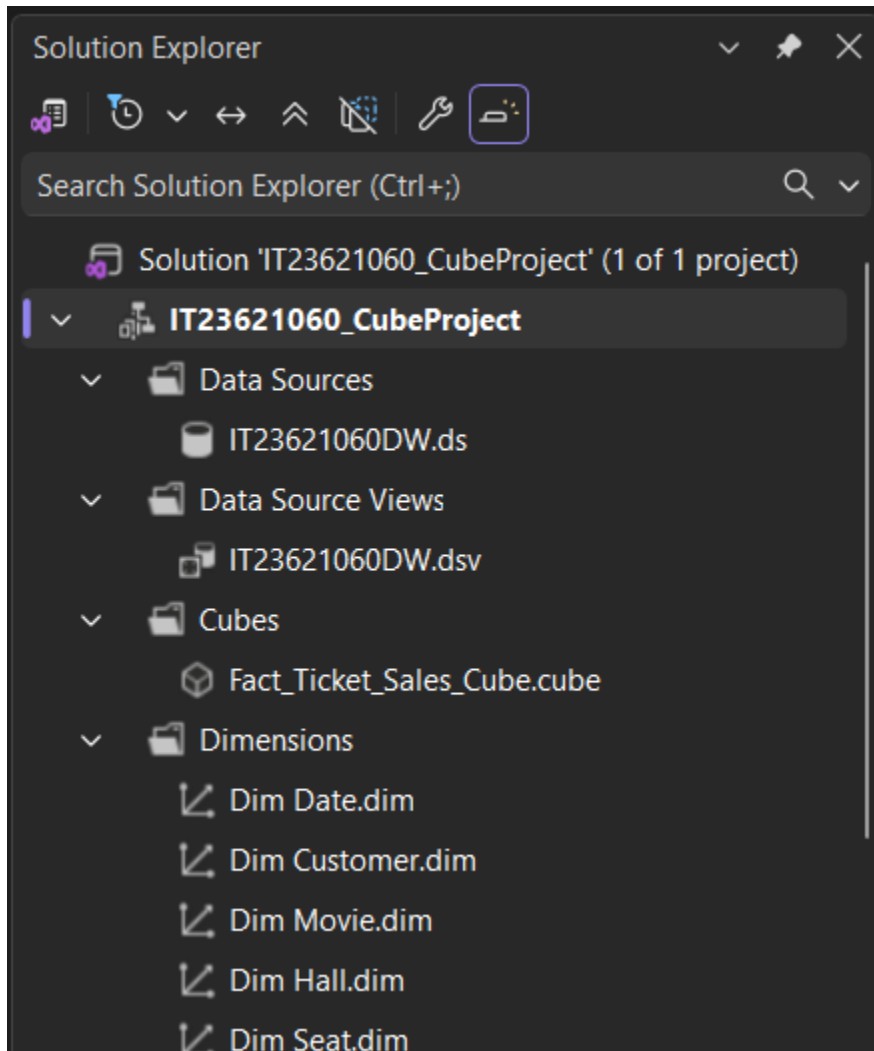
- Star schema of the cinema ticket sales data warehouse.



Step 2: SSAS Cube implementation

2.1. SSAS Cube implementation

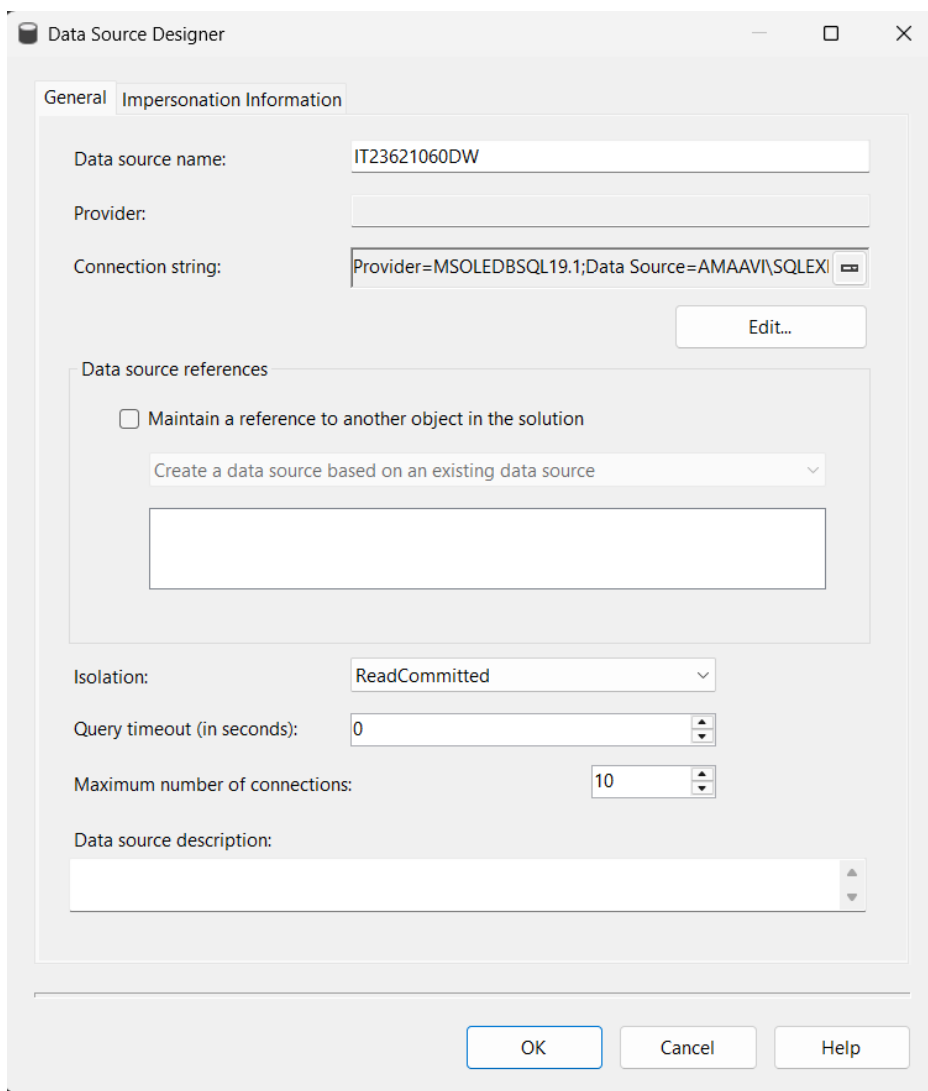
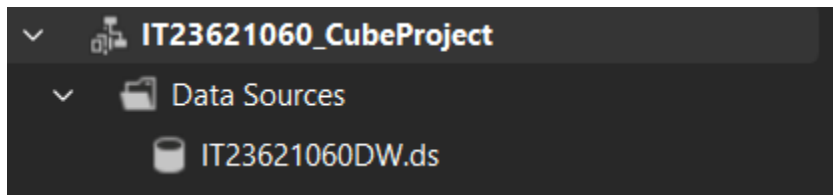
A new Analysis Services Multidimensional Project was created in Visual Studio. The SQL Server Analysis Services instance installed on the local machine was used as the deployment target.



Analysis Services Multidimensional Project created in Visual Studio.

2.2. Creating the Data Source

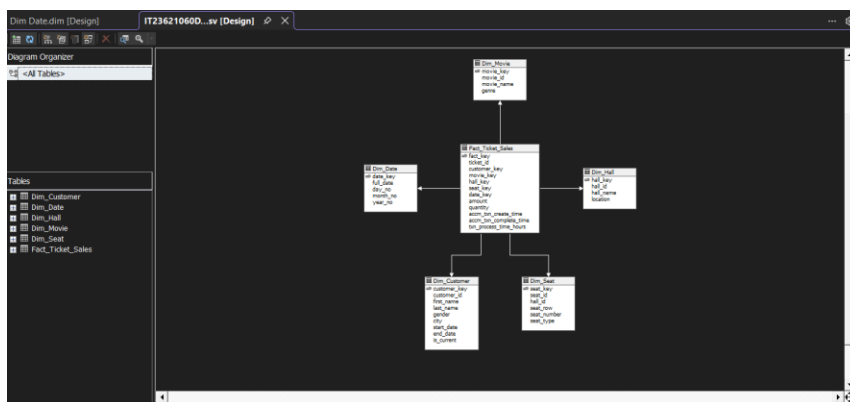
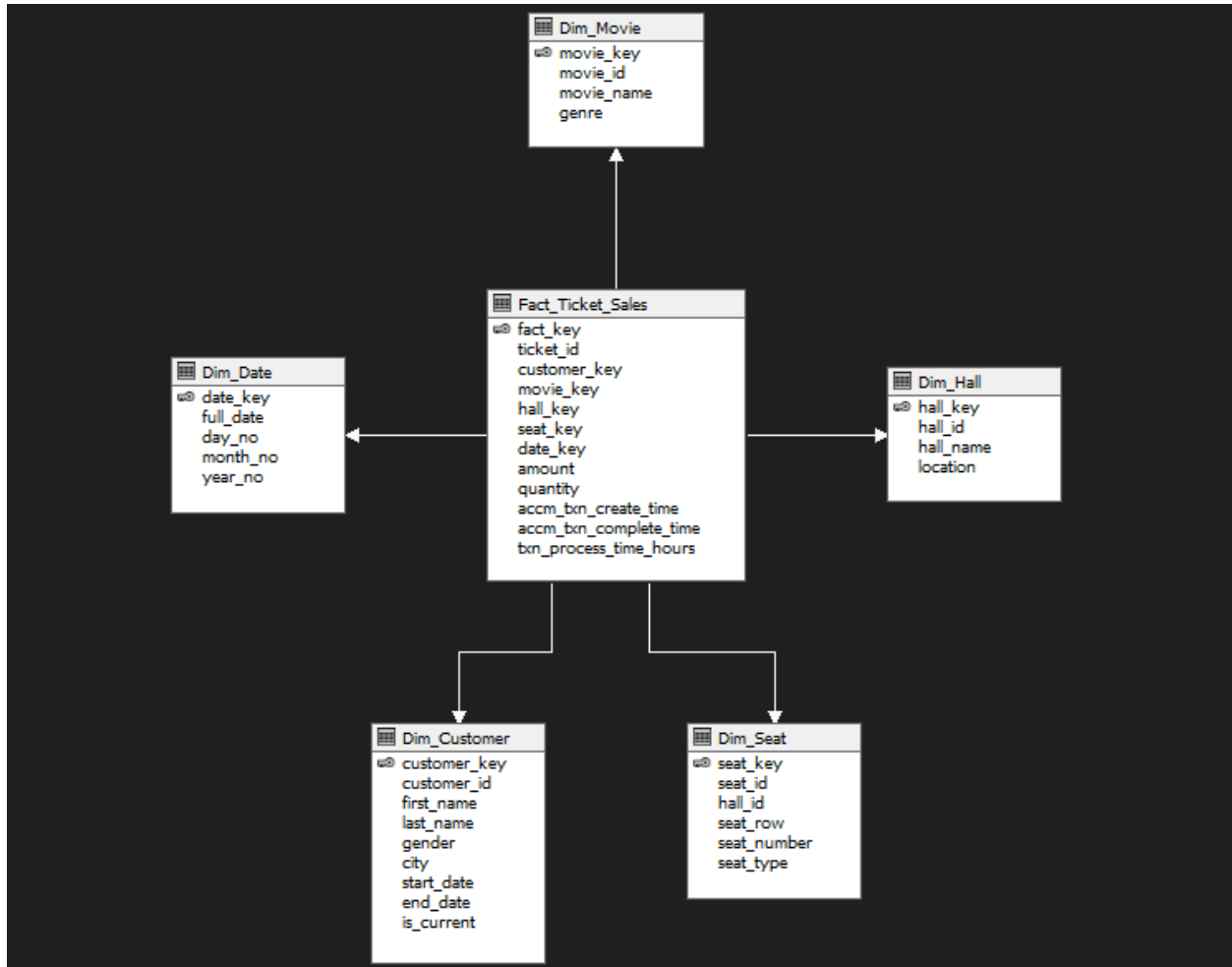
A data source was created by connecting the SSAS project to the SQL Server database engine containing the IT23621060DW warehouse. Windows Authentication was used and the existing warehouse database was selected.



Data source connection to the IT23621060DW warehouse database

2.3. Creating the Data Source View

A Data Source View was created using the fact and dimension tables from the warehouse. The fact table Fact_Ticket_Sales was linked with the dimensions through surrogate keys.

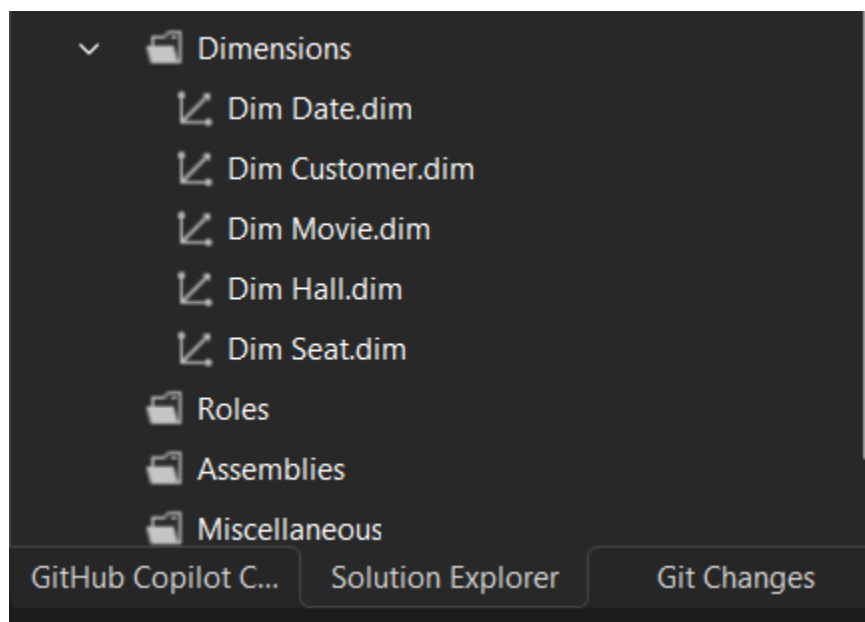


Data Source View showing the star schema relationships.

2.4. Creating dimensions

Dimensions were created from the existing dimension tables. User-friendly business attributes were selected for browsing, while technical ETL columns were excluded where not needed.

- Dim_Customer → customer details
- Dim_Movie → movie details
- Dim_Hall → hall details
- Dim_Seat → seat details
- Dim_Date → date analysis



Dimension Wizard

Specify Source Information

Select a data source and specify how the dimension is bound to it.



Data source view:

IT23621060DW

Main table:

Dim_Hall

Key columns:

hall_key

(Add key column)

hall_key

< Back

Next >

Finish >>|

Cancel

Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.



Available attributes:

☐ Attribute Name	☒ Enable Browsing	Attribute Type
☒ Seat Key	☒	Regular
☒ Seat Id	☒	Regular
☐ Hall Id	☐	Regular
☒ Seat Row	☒	Regular
☒ Seat Number	☒	Regular
☒ Seat Type	☒	Regular

< Back

Next >

Finish >>|

Cancel

Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.



Available attributes:

☐ Attribute Name	☒ Enable Browsing	Attribute Type
☒ Seat Key	☒	Regular
☒ Seat Id	☒	Regular
☐ Hall Id	☐	Regular
☒ Seat Row	☒	Regular
☒ Seat Number	☒	Regular
☒ Seat Type	☒	Regular

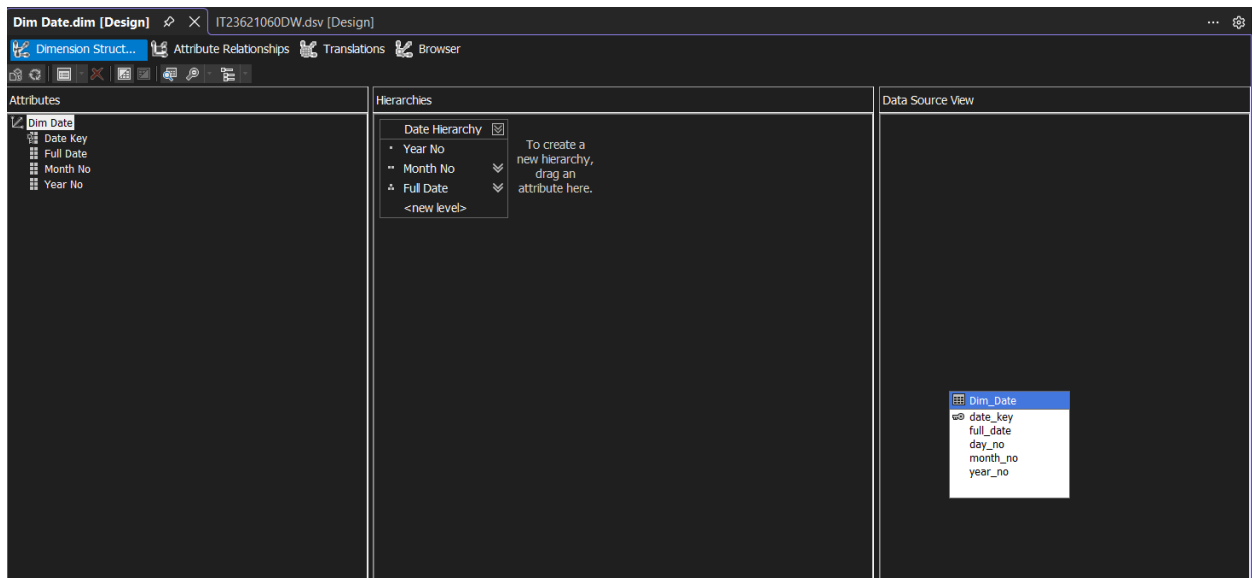
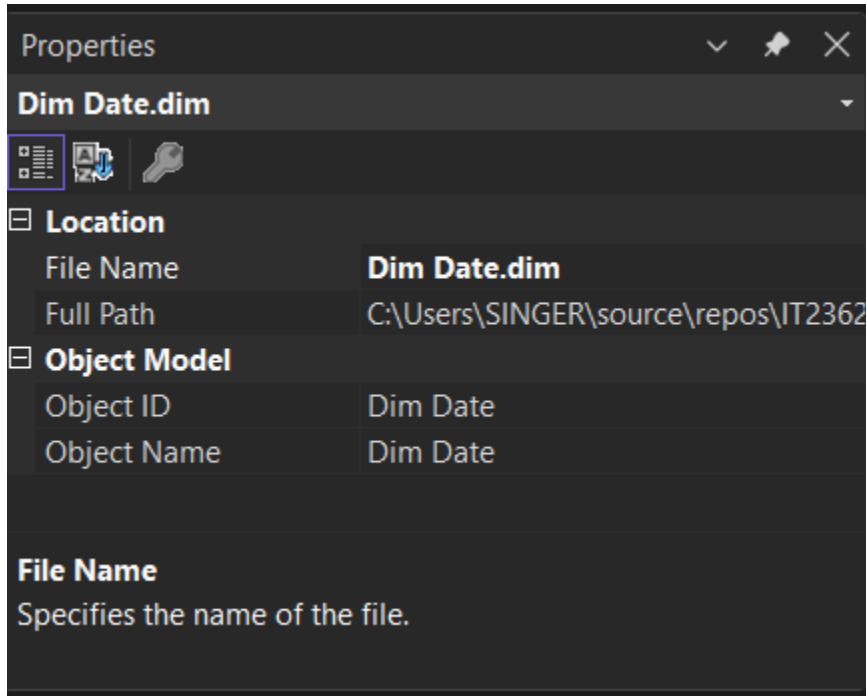
< Back

Next >

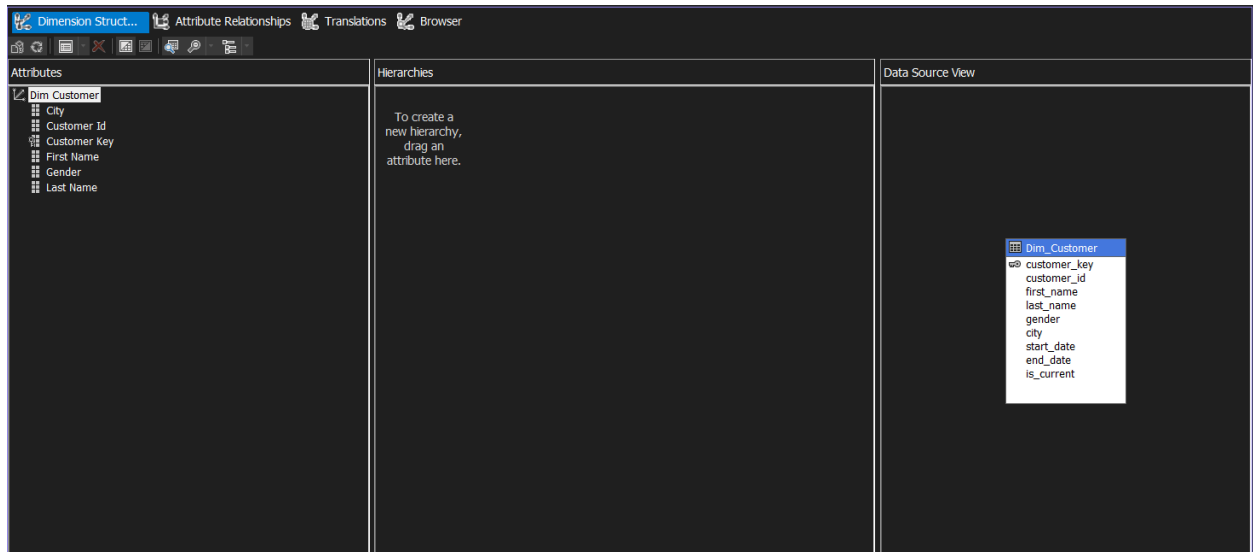
Finish >>|

Cancel

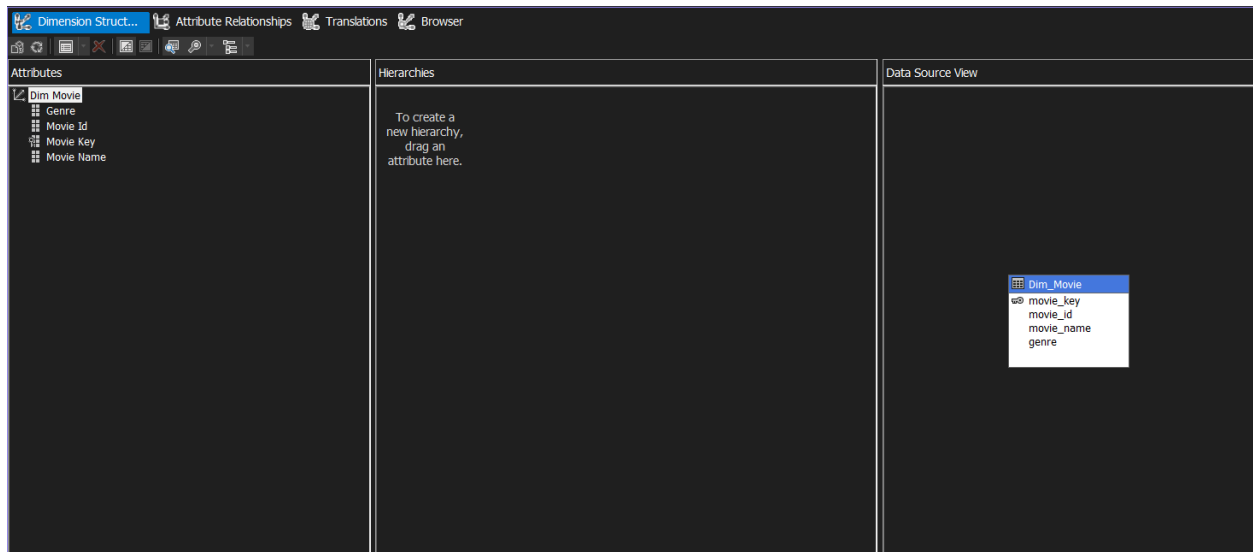
1. **Dim_Date dimension** - Date dimension created from Dim_Date.



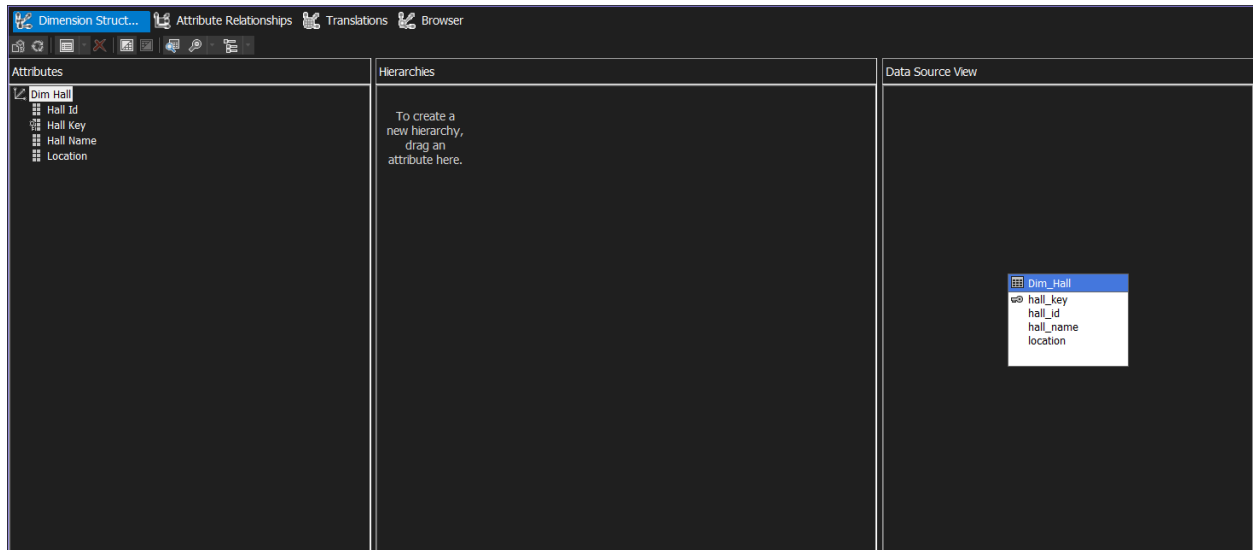
2. Dim_Customer dimension- Customer dimension created from Dim_Customer.



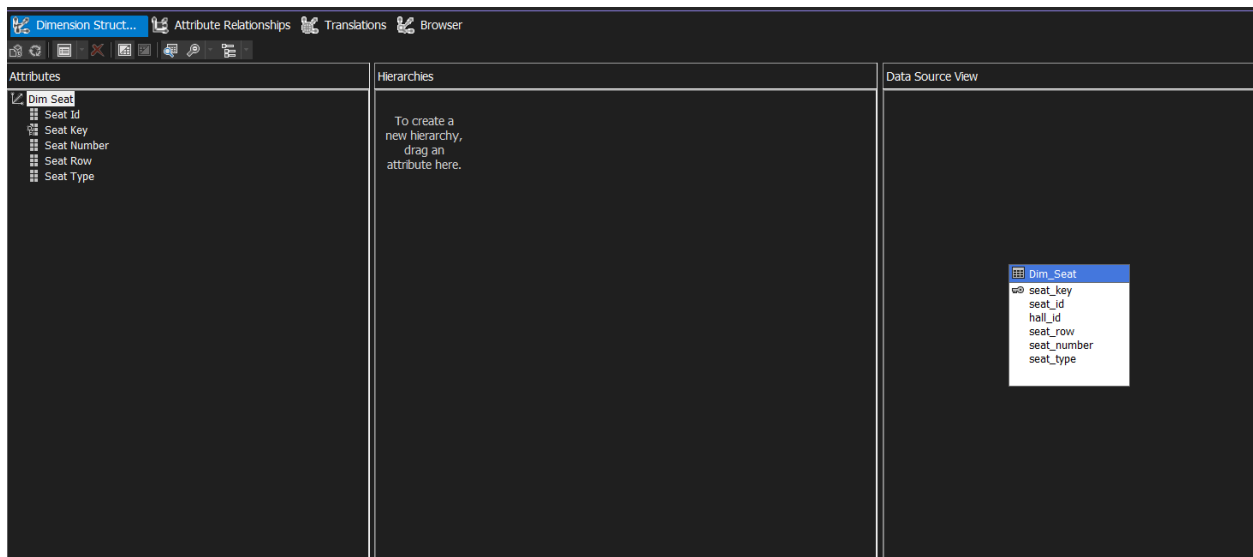
3. Dim_Movie dimension- Movie dimension created from Dim_Movie.



4. Dim_Hall dimension- Hall dimension created from Dim_Hall.

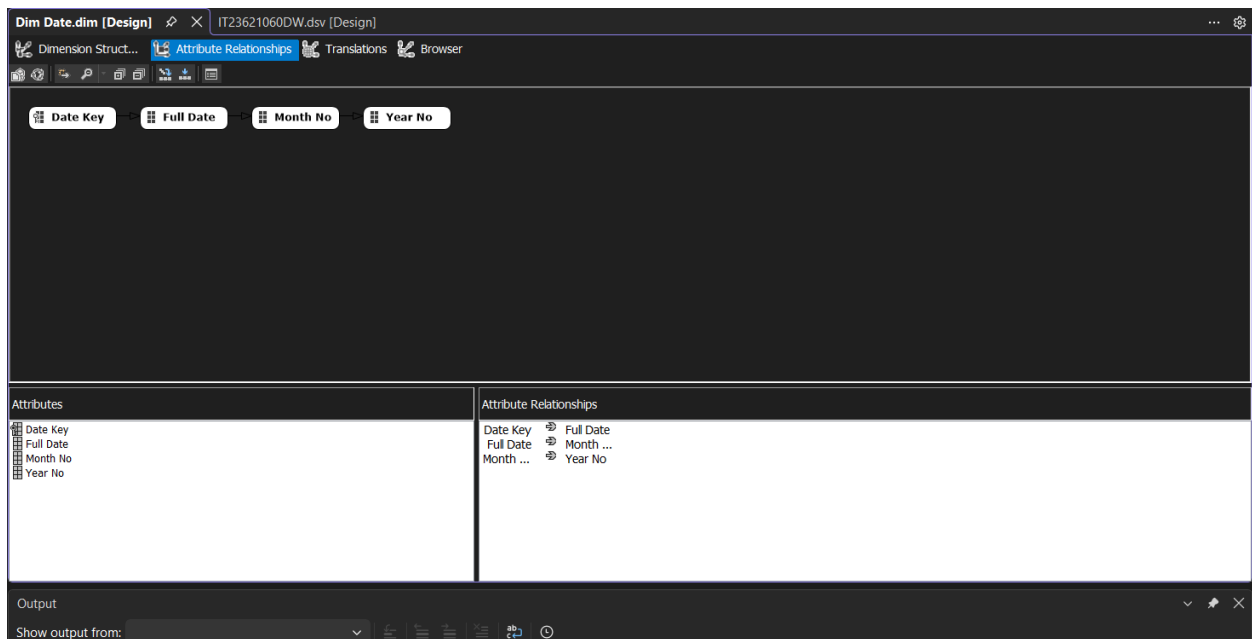
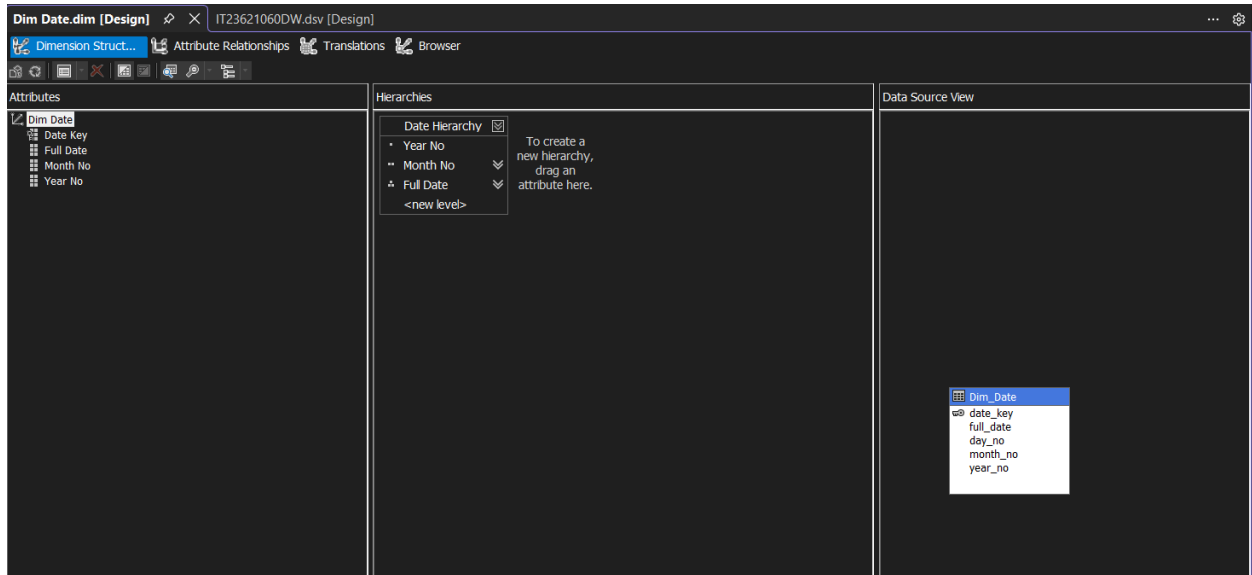


5. Dim_Seat dimension- Seat dimension created from Dim_Seat.



2.5. Creating the hierarchy

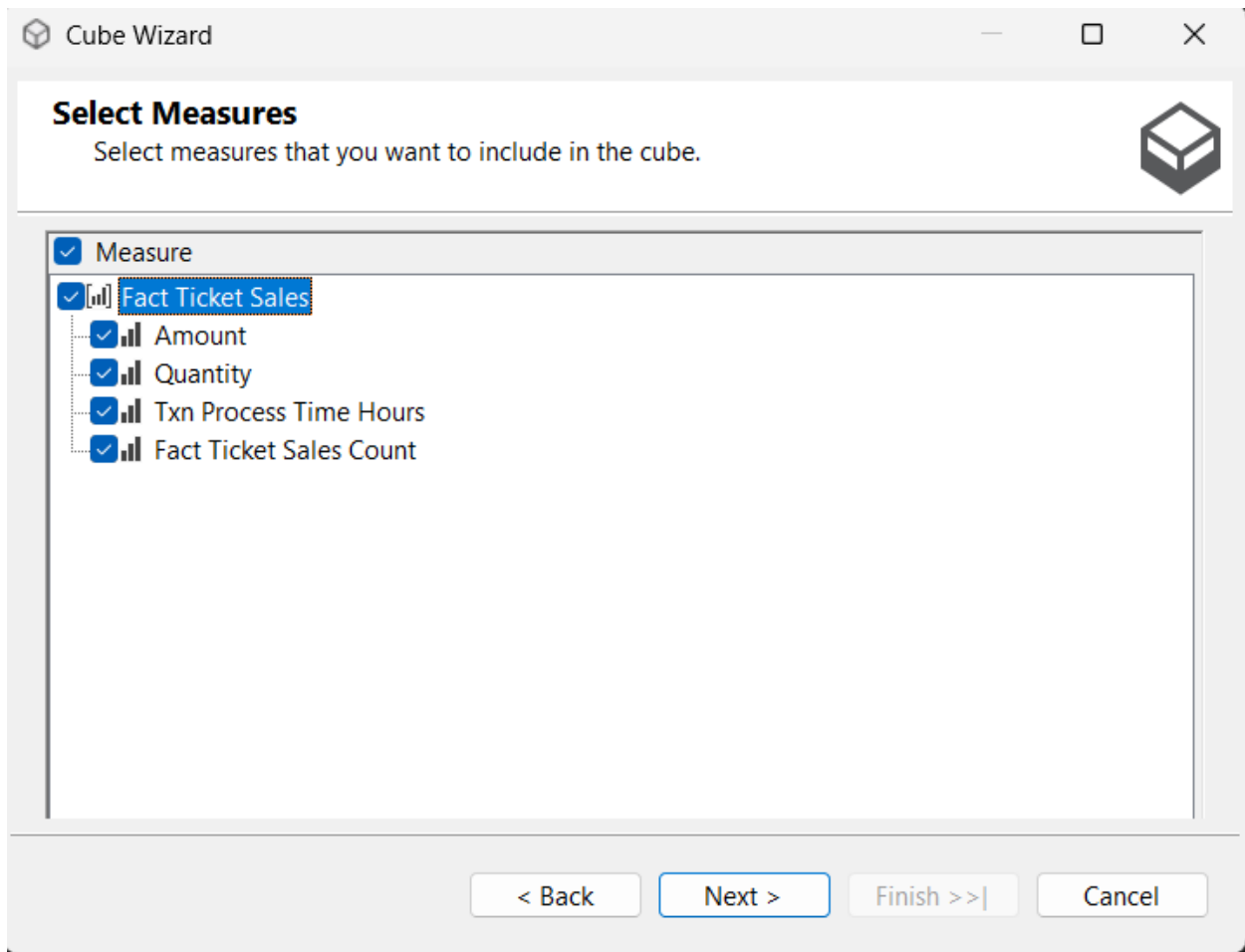
A date hierarchy was implemented in the Dim_Date dimension using Year No, Month No, and Full Date. This hierarchy supports hierarchical browsing and drill-down analysis in the cube.



Date hierarchy created in the Date dimension.

2.6. Creating the cube

The cube was created using Fact_Ticket_Sales as the measure group. Measures such as Total Sales, Total Tickets, Process Time Hours, and Transaction Count were selected. The dimensions Customer, Movie, Hall, Seat, and Date were added to the cube.



Cube Wizard

Select Existing Dimensions

Select existing dimensions to include in the cube.

- ☒ Dimension
 - ☒ Dim Date
 - ☒ Dim Customer
 - ☒ Dim Movie
 - ☒ Dim Hall
 - ☒ Dim Seat

< Back Next > Finish >> | Cancel

Cube Wizard

Completing the Wizard

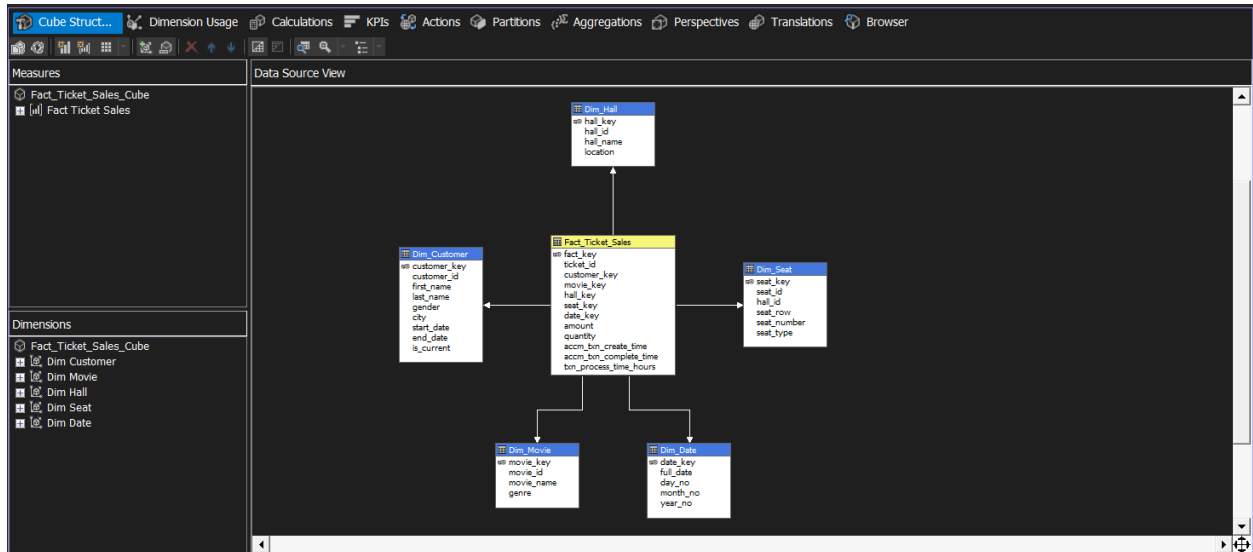
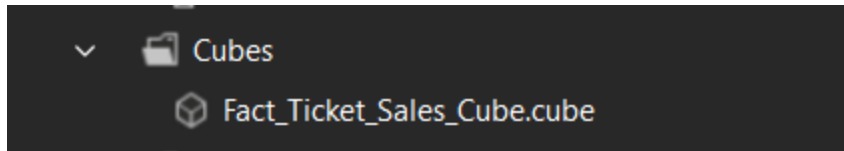
Name the cube, review its structure, and then click Finish to save the cube.

Cube name:
IT23621060DW

Preview:

- Total Sales
- Total Tickets
- Process Time Hours
- Transaction Count
- Dimensions
 - Dim Date
 - Dim Customer
 - Dim Movie
 - Dim Hall
 - Dim Seat
 - Fact Ticket Sales

< Back Next > Finish Cancel

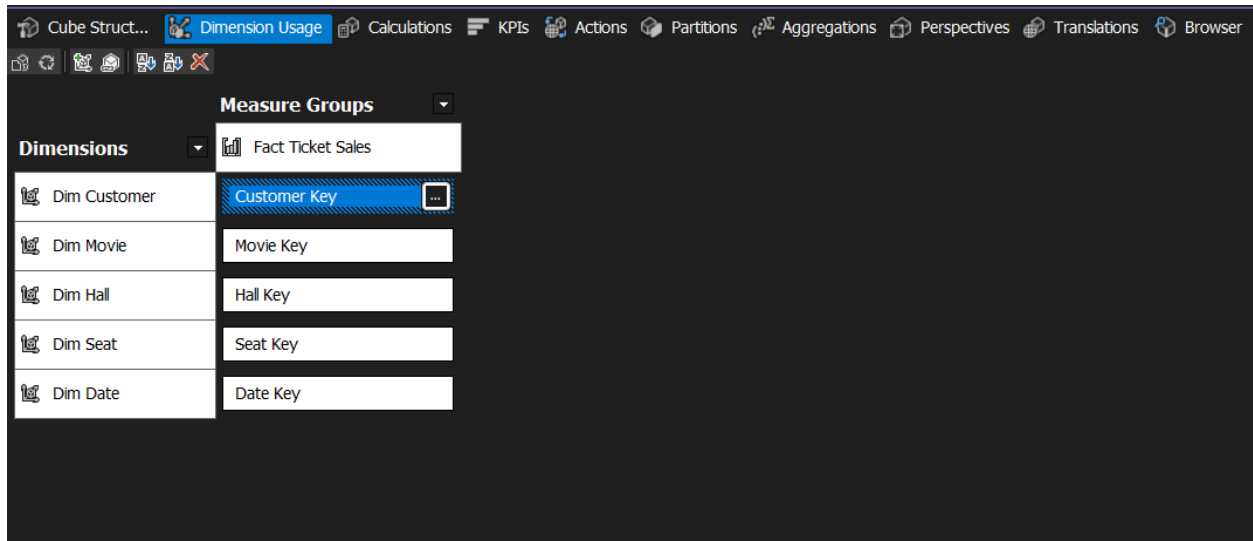


Cube created using Fact_Ticket_Sales as the measure group.

The cube was designed using Fact_Ticket_Sales as the central measure group because it contains the numeric measures required for analysis. The dimensions were added to support slicing and drilling across customer, movie, hall, seat, and date perspectives.

2.7. Dimension usage

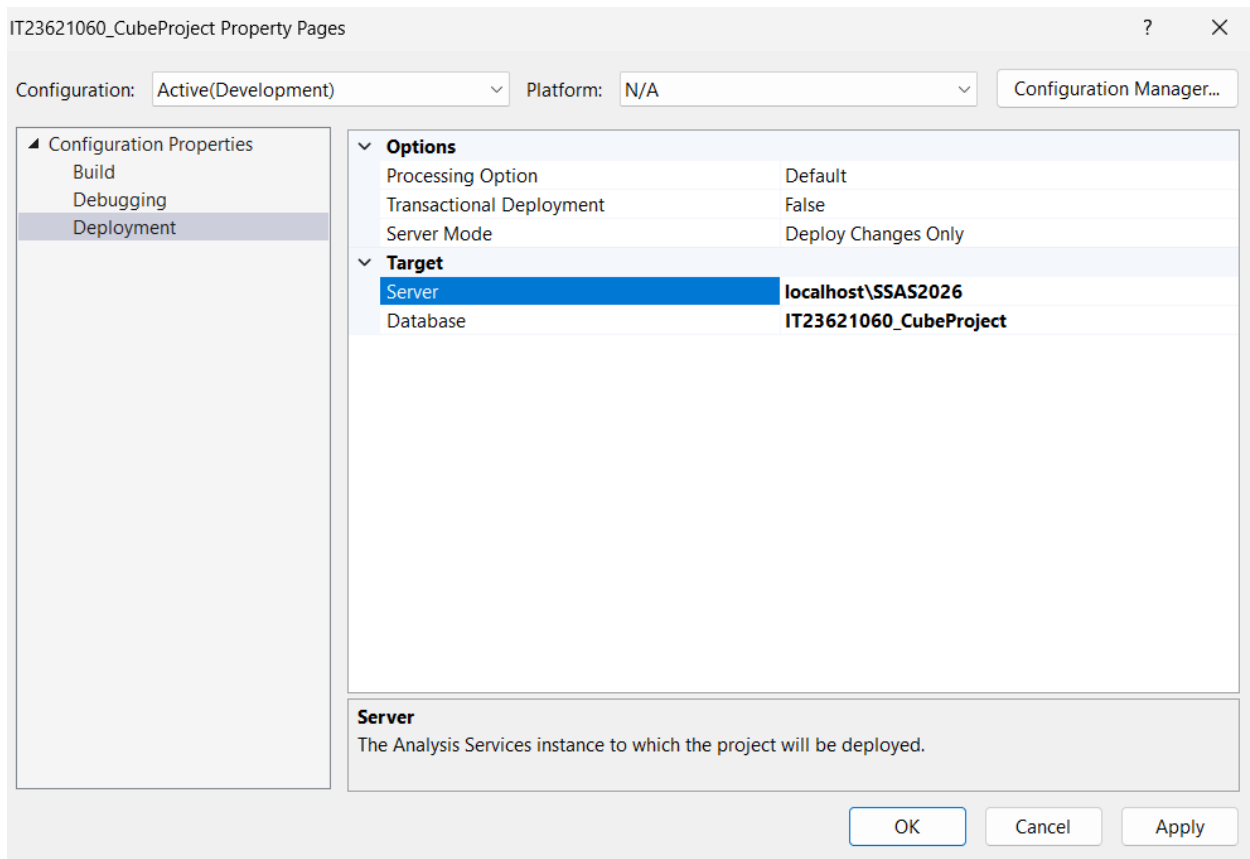
The Dimension Usage tab was checked to ensure that all dimensions were correctly mapped to the fact table using their surrogate keys. This confirmed the logical relationship between the measure group and each dimension.



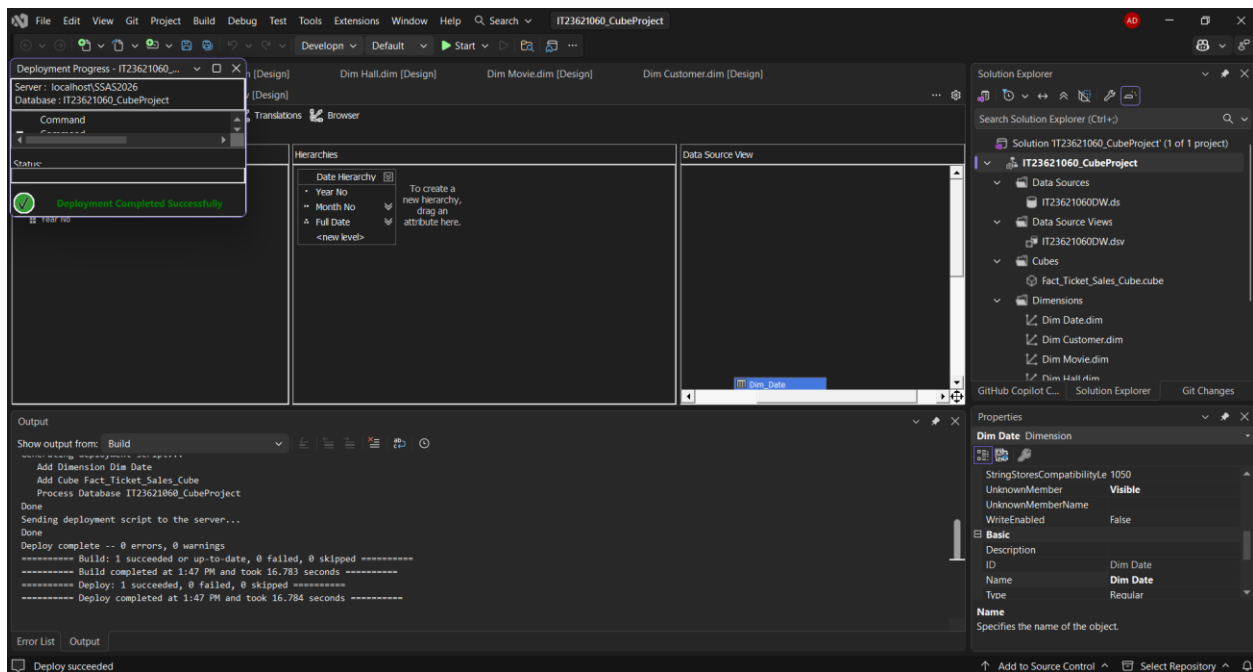
Dimension Usage showing correct relationships between fact and dimensions.

2.8. Deployment and processing

The SSAS project was deployed to the local Analysis Services instance localhost\SSAS2026. After deployment, the database was processed successfully so that the cube could be browsed.



Deployment target set to the local SSAS instance.



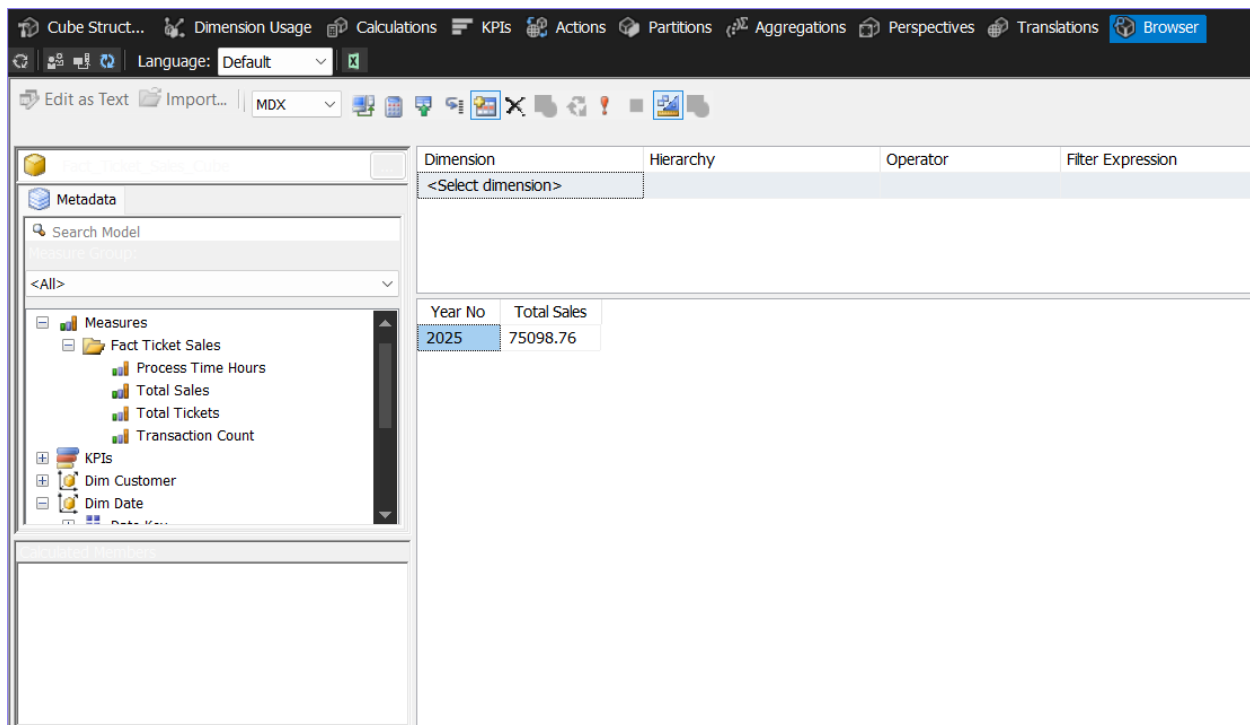
Successful deployment and processing of the cube.

Deployment published the cube structure to the Analysis Services server, while processing loaded the warehouse data into the cube so that it could be queried from the Browser tab and external tools such as Excel.

2.9. Browsing the cube

The Browser tab was used to validate the cube. Measures and dimensions were dragged into the query area to confirm that the cube returned correct analytical results.

These browser results confirmed that the measures, dimensions, and hierarchy were functioning correctly in the deployed cube.



The screenshot shows the 'Browser' tab of a cube management tool. The interface includes a top menu bar with options like 'Cube Structure', 'Dimension Usage', 'Calculations', 'KPIs', 'Actions', 'Partitions', 'Aggregations', 'Perspectives', 'Translations', and 'Browser'. Below the menu is a toolbar with icons for 'Edit as Text', 'Import...', and 'MDX'. The main workspace is divided into three sections: a left sidebar for 'Metadata' with a search bar and a tree view of measures and dimensions; a top right section for defining the query with columns for 'Dimension', 'Hierarchy', 'Operator', and 'Filter Expression'; and a bottom right section displaying the query results in a table.

Year No	Total Sales
2025	75098.76

Cube browser showing Total Sales by Year.

The screenshot shows the Cube browser interface. On the left, the 'Fact_Ticket_Sales_Cube' is selected, showing a tree view of measures including 'Total Sales' and 'Total Tickets'. The main area displays a table with the following data:

Year No	Movie Name	Total Sales	Total Tickets
2025	Action Blast	15794.91	2604
2025	Comedy Ni...	14880.96	2496
2025	Drama Life	14752.68	2520
2025	Horror House	15889.56	2586
2025	SciFi World	13780.65	2244

Cube browser showing Total Sales and Total Tickets by Movie.

The screenshot shows the Cube browser interface. On the left, the 'Fact_Ticket_Sales_Cube' is selected, showing a tree view of measures including 'Total Sales'. The main area displays a table with the following data:

Year No	Month No	Full Date	Total Sales
2025	1	2025-0...	217.92
2025	1	2025-0...	197.34
2025	1	2025-0...	202.17
2025	1	2025-0...	217.5
2025	1	2025-0...	182.28
2025	1	2025-0...	210.93
2025	1	2025-0...	214.62
2025	1	2025-0...	205.71
2025	1	2025-0...	196.98
2025	1	2025-0...	195.87
2025	2	2025-0...	205.23
2025	2	2025-0...	236.22
2025	2	2025-0...	234.33
2025	2	2025-0...	186.3
2025	2	2025-0...	228.99
2025	2	2025-0...	175.68

Cube browser showing Date Hierarchy with Total Sales.

Step 3 — Excel OLAP operations

Excel was connected to the deployed SSAS cube using the Analysis Services connection option. A PivotTable was created in a new worksheet for OLAP analysis.

The screenshot shows the 'Data Connection Wizard' dialog box, specifically the 'Connect to Database Server' step. The title bar reads 'Data Connection Wizard' with a help icon (?) and a close icon (X). The main heading is 'Connect to Database Server', followed by the instruction 'Enter the information required to connect to the database server.'.

Step 1, 'Server name:', has a text box containing 'localhost\SSAS2026'.

Step 2, 'Log on credentials', has two radio button options: 'Use Windows Authentication' (which is selected) and 'Use the following User Name and Password'. Below these options are two text boxes: 'User Name:' and 'Password:', both of which are currently empty.

At the bottom of the dialog are four buttons: 'Cancel', '< Back', 'Next >' (which is highlighted with a blue border), and 'Finish'.

Data Connection Wizard

?

×

Select Database and Table

Select the Database and Table/Cube which contains the data you want.

Select the database that contains the data you want:

IT23621060_CubeProject

▼

☒ Connect to a specific cube or table:

Name	Description	Modified	Created	Type
 Fact_Ticket_Sales_Cube		4/1/2026 1:47:57 PM		CUBE

Cancel

< Back

Next >

Finish

Excel connected to the deployed SSAS cube using an Analysis Services connection.

3.1. Roll-up

Roll-up was demonstrated by summarizing Total Sales at the Year level using a PivotTable.

The screenshot displays the Microsoft Excel interface. The PivotTable is located in the range A4:B6. The PivotTable Fields task pane is open on the right side of the window.

Row Labels	Total Sales
# 2025	75098.76
Grand Total	75098.76

PivotTable Fields

Choose fields to add to report:

- ☒ Fact Ticket Sales
 - ☐ Process Time Hours
 - ☒ Total Sales
 - ☐ Total Tickets
 - ☐ Transaction Count
- ☒ Dim Customer
 - ☐ City
 - ☐ Customer Id

Drag fields between areas below:

Filters	Columns

Rows	Values
Date Hierarchy	Total Sales

☐ Defer Layout Update Update

Roll-up operation showing Total Sales by Years

3.2. Drill-down

Drill-down was demonstrated using the Date Hierarchy, allowing navigation from Year level to Month and Full Date levels.

The screenshot displays an Excel PivotTable with the following data:

Row Labels	Total Sales
2025	196.98
2025-01-01	198.84
2025-01-02	197.73
2025-01-03	218.46
2025-01-04	244.98
2025-01-05	205.29
2025-01-06	207
2025-01-07	267.75
2025-01-08	232.35
2025-01-09	164.49
2025-01-10	193.35
2025-01-11	254.82
2025-01-12	162.9
2025-01-13	210.51
2025-01-14	212.52
2025-01-15	213.78
2025-01-16	193.35
2025-01-17	204.72
2025-01-18	193.71
2025-01-19	220.53
2025-01-20	184.35
2025-01-21	235.23
2025-01-22	217.92
2025-01-23	197.34
2025-01-24	202.17
2025-01-25	217.5
2025-01-26	182.28
2025-01-27	210.93
2025-01-28	214.62
2025-01-29	205.71
2025-01-30	196.98

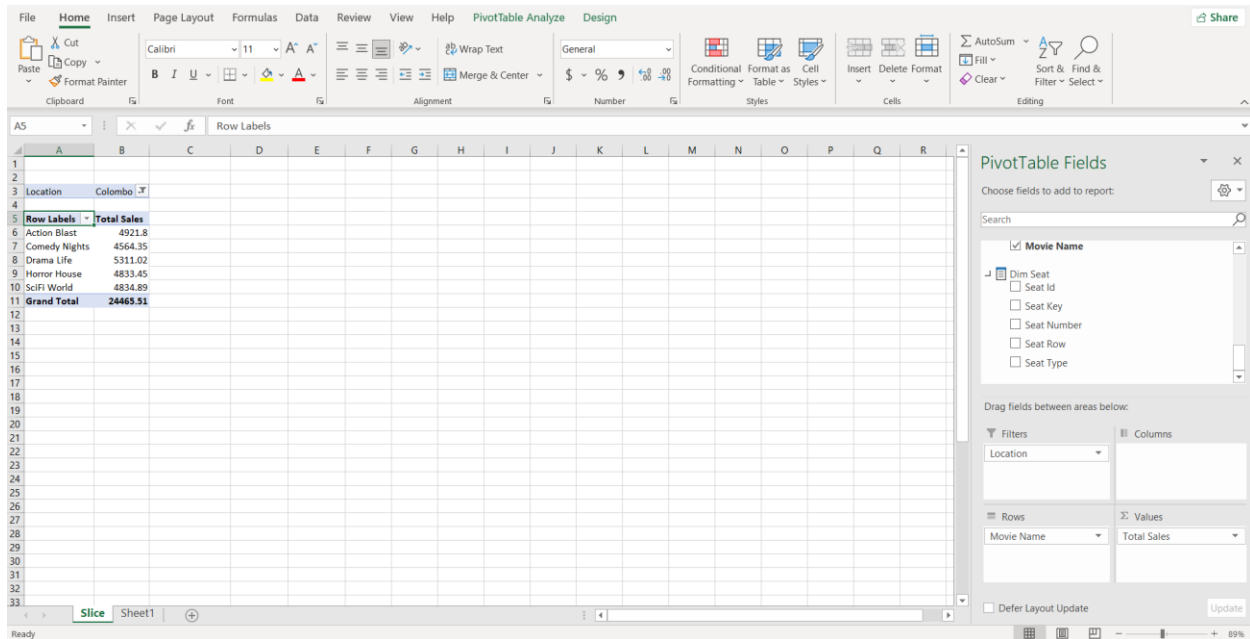
The PivotTable Fields task pane on the right shows the hierarchy structure:

- Choose fields to add to report:
 - ☐ Total Tickets
 - ☐ Transaction Count
 - ☒ Dim Customer
 - ☐ City
 - ☐ Customer Id
 - ☐ Customer Key
 - ☐ First Name
- Drag fields between areas below:
 - Filters: (empty)
 - Columns: (empty)
 - Rows: Date Hierarchy
 - Values: Total Sales
- Defer Layout Update: ☐
- Update: [button]

Drill-down operation using the Date Hierarchy.

3.3. Slice

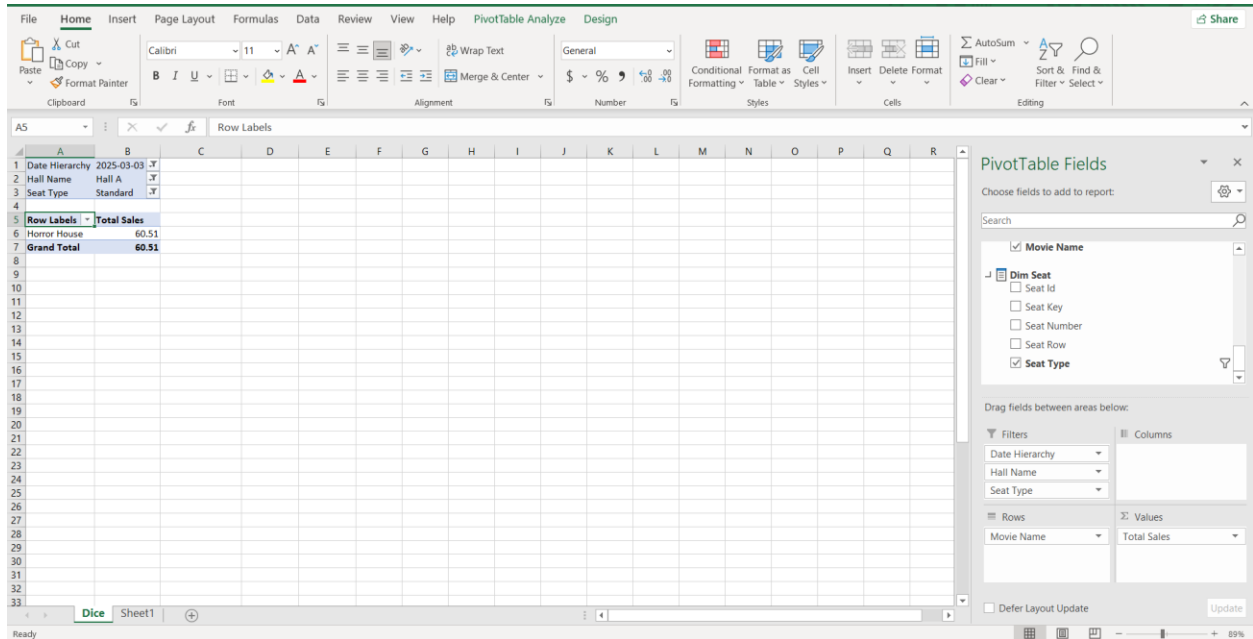
Slice was demonstrated by applying a filter on a single dimension value, such as one hall or one location, to analyze a smaller subset of the cube.



Slice operation using a single filter value.

3.4. Dice

Dice was demonstrated by applying multiple filters simultaneously, such as Year and Hall, to analyze a more specific subset of the data.



Dice operation using multiple filter values.

3.5. Pivot

Pivot was demonstrated by arranging Movie Name in rows and Hall Name in columns with Total Sales as the value, then switching the arrangement to view the same data from a different perspective

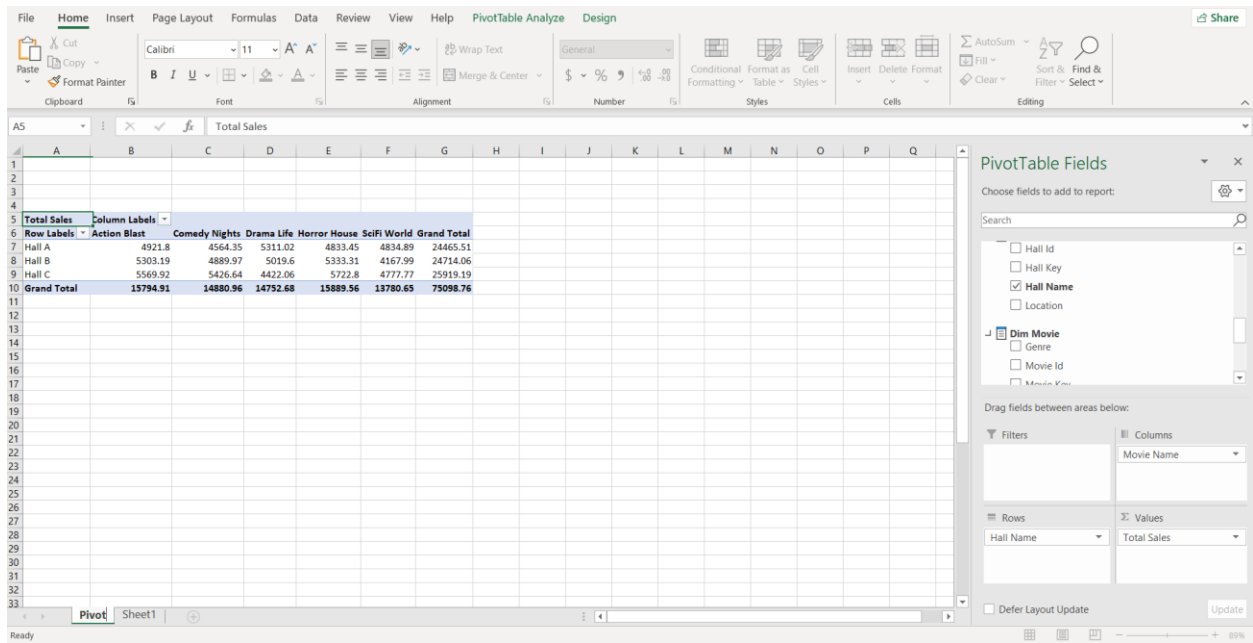
The screenshot shows an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is titled 'Total Sales' and is located in the range A5:E12. The task pane is on the right side of the screen, showing the fields available for the PivotTable.

PivotTable Data:

Row Labels	Hall A	Hall B	Hall C	Grand Total
Action Blast	4921.8	5303.19	5569.92	15794.91
Comedy Nights	4564.35	4889.97	5426.64	14880.96
Drama Life	5311.02	5019.6	4422.06	14752.68
Horror House	4833.45	5333.31	5722.8	15889.56
SciFi World	4834.89	4167.99	4777.77	13780.65
Grand Total	24465.51	24714.06	25919.19	75098.76

PivotTable Fields Task Pane:

- Choose fields to add to report:
 - ☐ Hall Id
 - ☐ Hall Key
 - ☒ Hall Name
 - ☐ Location
- ☒ Dim Movie
 - ☐ Genre
 - ☐ Movie Id
 - ☐ Movie Key
- Drag fields between areas below:
 - Filters: (empty)
 - Columns: Hall Name
 - Rows: Movie Name
 - Values: Total Sales
- Defer Layout Update: ☐ Update



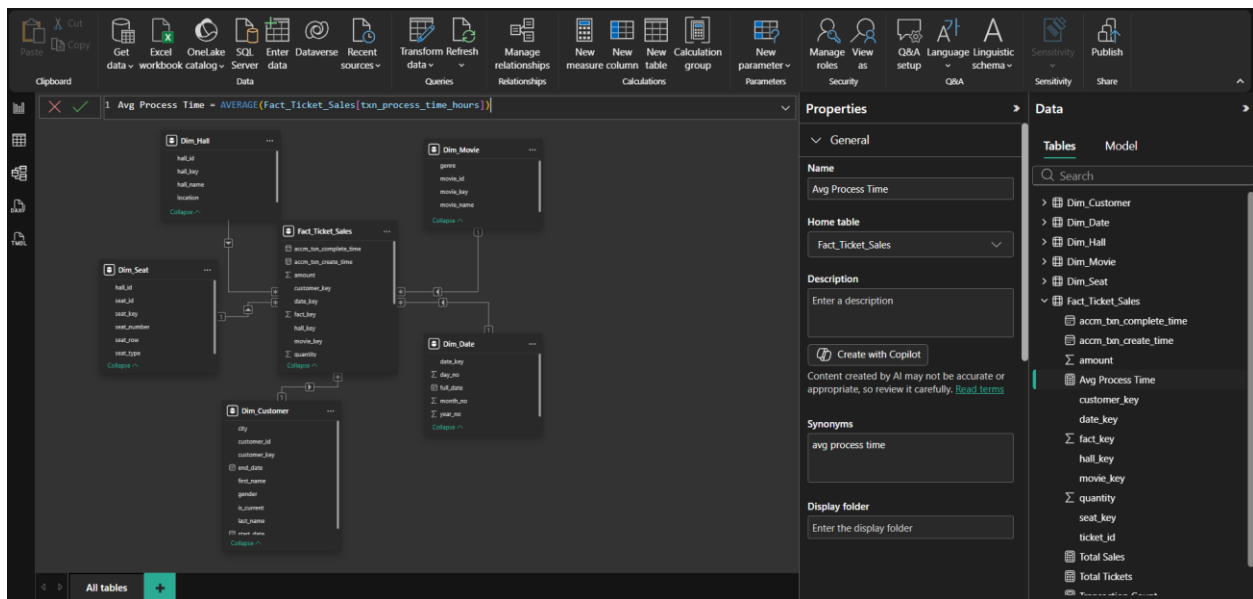
Pivot operation showing Total Sales by Movie and Hall

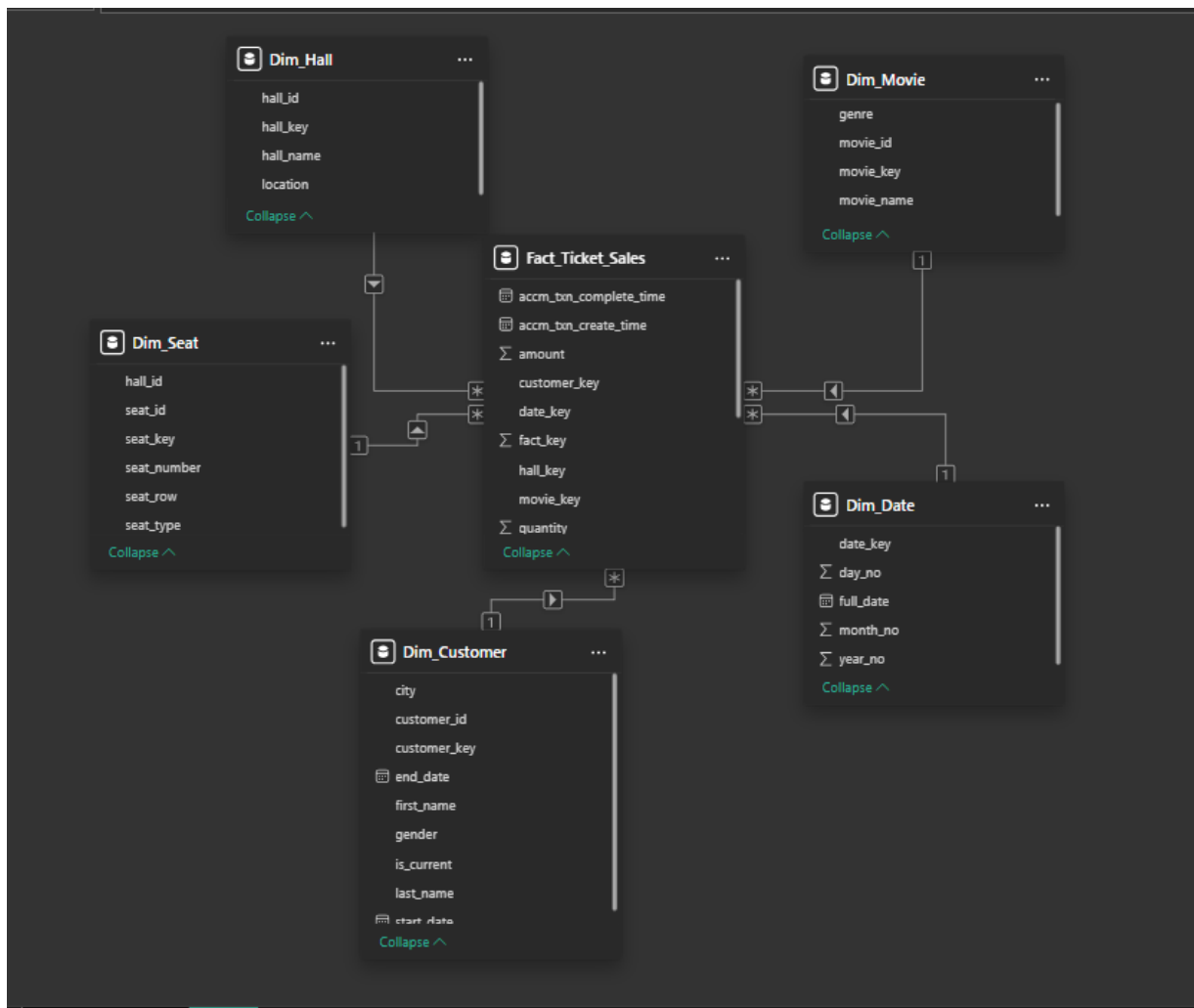
These PivotTable views confirmed that the cube could successfully support all required OLAP operations in Excel.

Step 4 — Power BI reports

4.1. Loading the warehouse into Power BI

Power BI Desktop was connected to the IT23621060DW database using SQL Server Import mode. After loading the fact and dimension tables, the model relationships were verified and DAX measures were created for reporting. The following report pages were then developed according to the assignment requirements.





Power BI data model with fact and dimension relationships.

4.2. Measures created in Power BI

The following DAX measures were created for reporting: Total Sales, Total Tickets, Transaction Count, and Average Process Time.



```
1 Total Sales = SUM(Fact_Ticket_Sales[amount])
```



```
1 Total Tickets = SUM(Fact_Ticket_Sales[quantity])
```



```
1 Transaction Count = COUNT(Fact_Ticket_Sales[fact_key])
```



```
1 Avg Process Time = AVERAGE(Fact_Ticket_Sales[txn_process_time_hours])
```

DAX measures created in Power BI

4.3. Report 1 – Matrix

A Matrix visual was created to display detailed tabular data with row and column groupings. Movie Genre and Movie Name were used in rows, while Year and Month were used in columns. Total Sales and Total Tickets were used as values.

year_no	2025	Total	Total	Total
genre	Total Sales	Total Tickets	Total Sales	Total Tickets
Action	15,794.91	2604	15,794.91	2604
Action Blast	15,794.91	2604	15,794.91	2604
Comedy	14,880.96	2496	14,880.96	2496
Comedy Nights	14,880.96	2496	14,880.96	2496
Drama	14,752.68	2520	14,752.68	2520
Drama Life	14,752.68	2520	14,752.68	2520
Horror	15,889.56	2586	15,889.56	2586
Horror House	15,889.56	2586	15,889.56	2586
Sci-Fi	13,780.65	2244	13,780.65	2244
SciFi World	13,780.65	2244	13,780.65	2244
Total	75,098.76	12450	75,098.76	12450

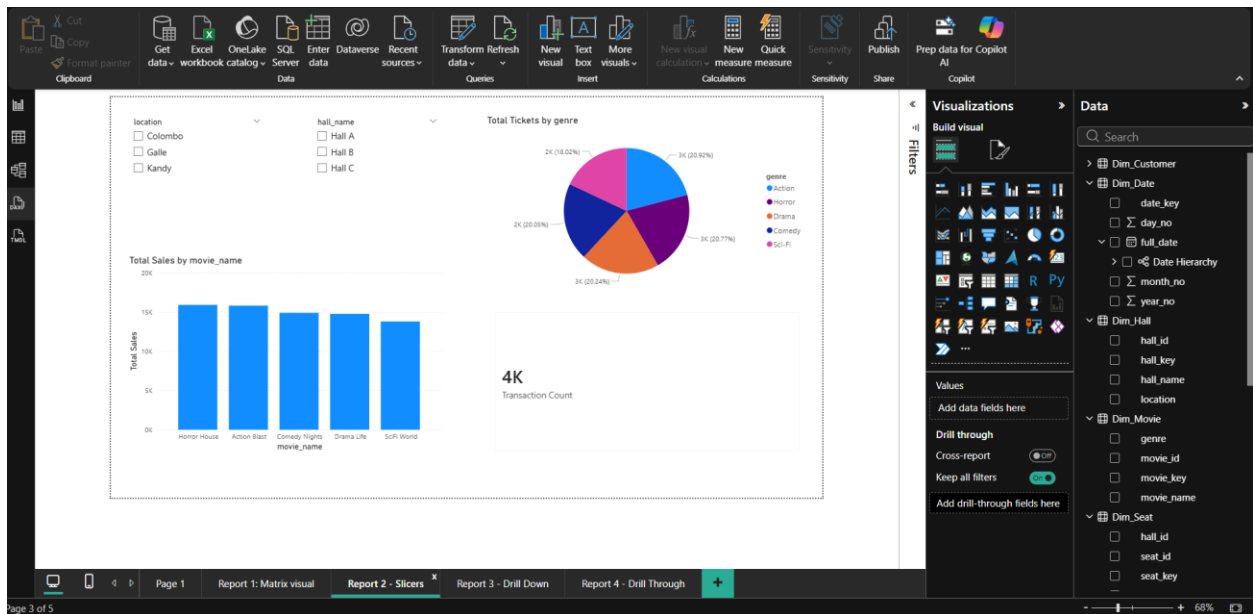
year_no	2025	Total		
genre	Total Sales	Total Tickets	Total Sales	Total Tickets
[-] Action	15,794.91	2604	15,794.91	2604
Action Blast	15,794.91	2604	15,794.91	2604
[-] Comedy	14,880.96	2496	14,880.96	2496
Comedy Nights	14,880.96	2496	14,880.96	2496
[-] Drama	14,752.68	2520	14,752.68	2520
Drama Life	14,752.68	2520	14,752.68	2520
[-] Horror	15,889.56	2586	15,889.56	2586
Horror House	15,889.56	2586	15,889.56	2586
[-] Sci-Fi	13,780.65	2244	13,780.65	2244
SciFi World	13,780.65	2244	13,780.65	2244
Total	75,098.76	12450	75,098.76	12450

Matrix visual displaying grouped movie sales data.

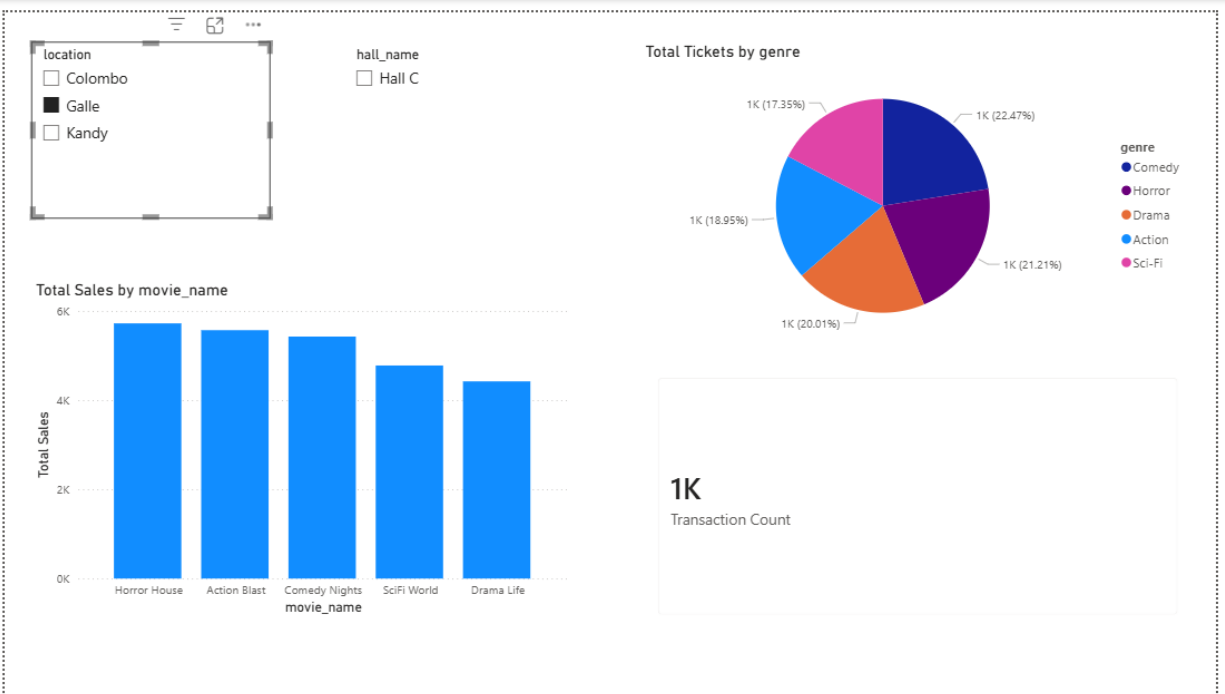
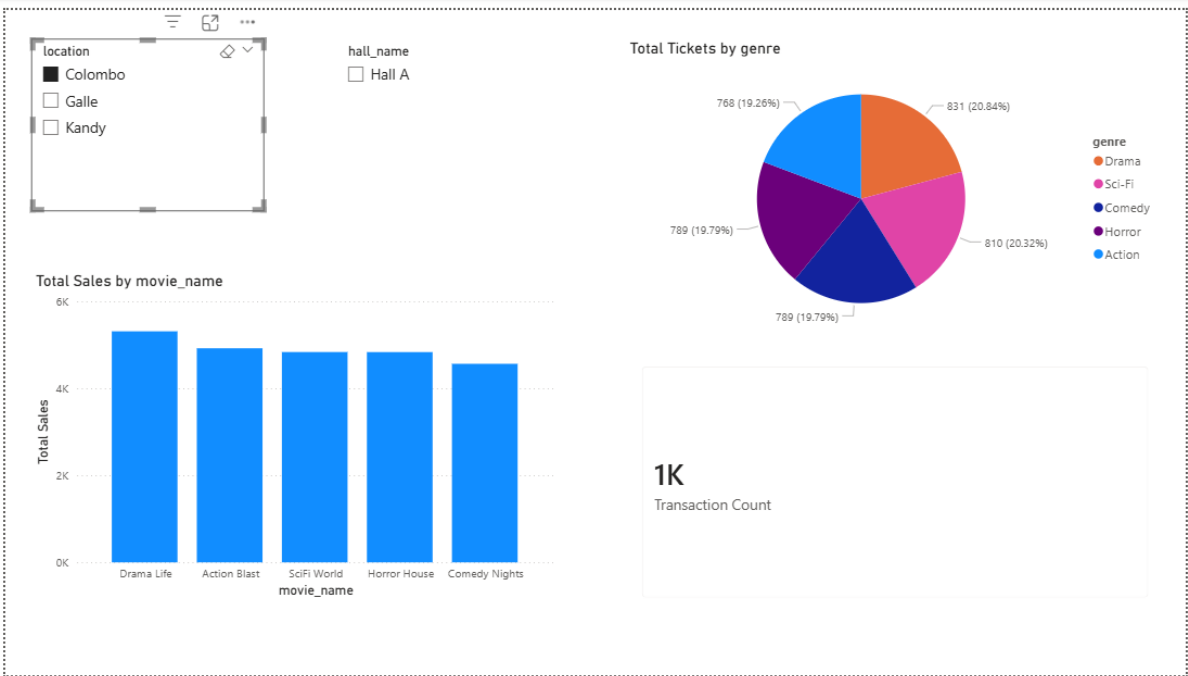
This report provides a tabular view of sales and ticket quantities grouped by movie and time period, making it useful for comparing performance across genres and years.

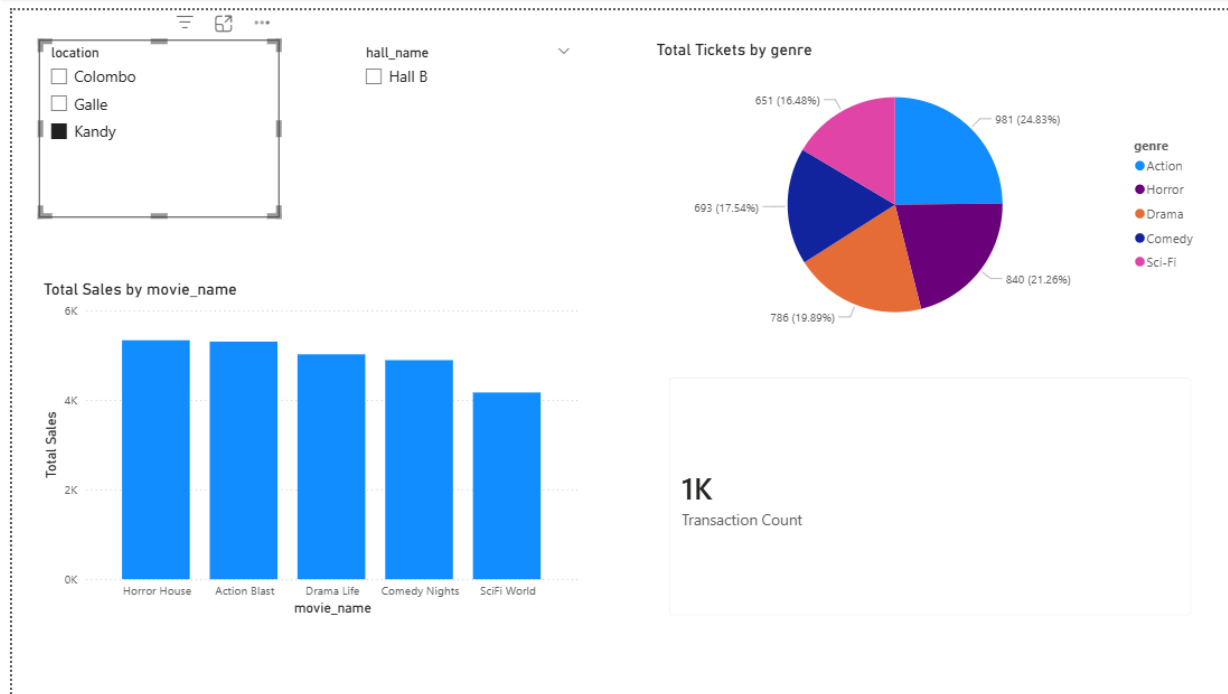
4.4. Report 2 - Multiple slicers with cascading filters

Two slicers were created using location and hall_name. The selection in the first slicer filtered the available values in the second slicer, demonstrating cascading filtering. Additional visuals were added to show sales, ticket distribution, and transaction counts.



Cascading slicers using Location and Hall Name.





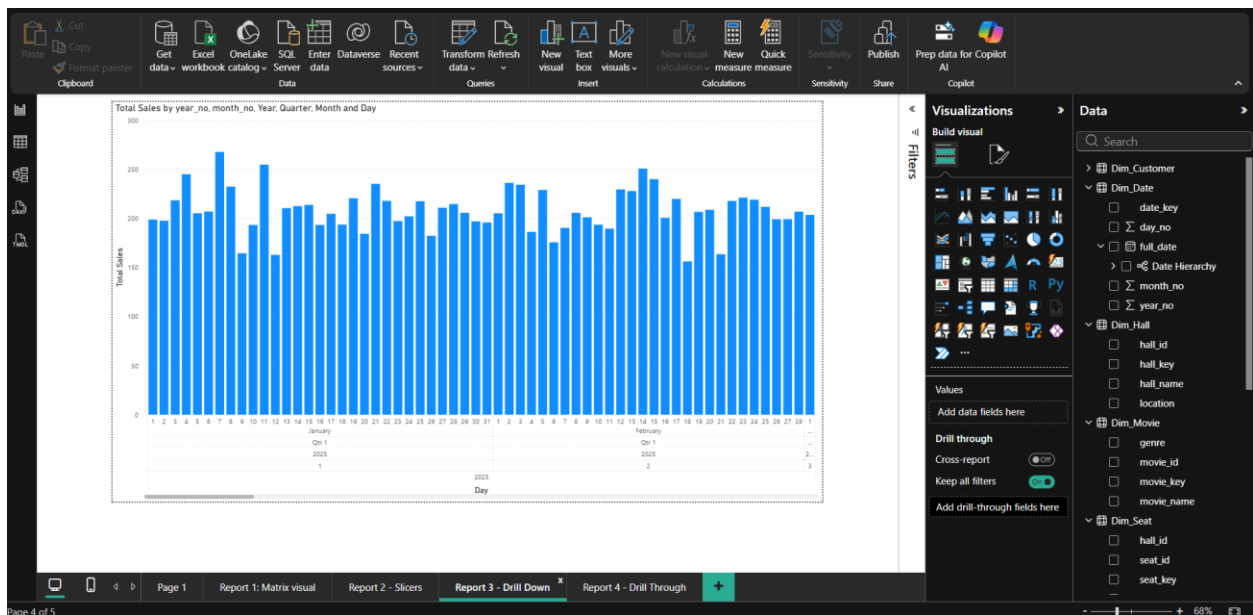
Visuals updated dynamically based on slicer selections.

The first slicer limits the values available in the second slicer, which demonstrates cascading filtering. The report also includes multiple visuals to display changes in sales, tickets, and transaction counts based on the selected filters.

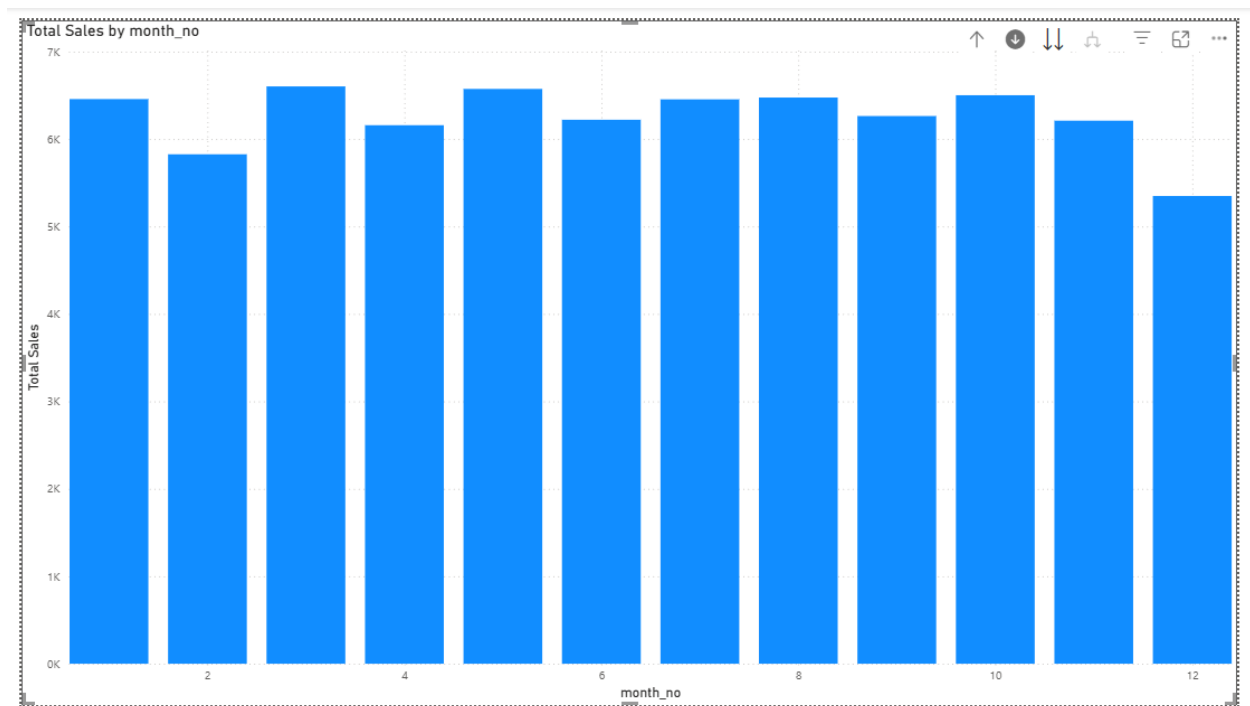
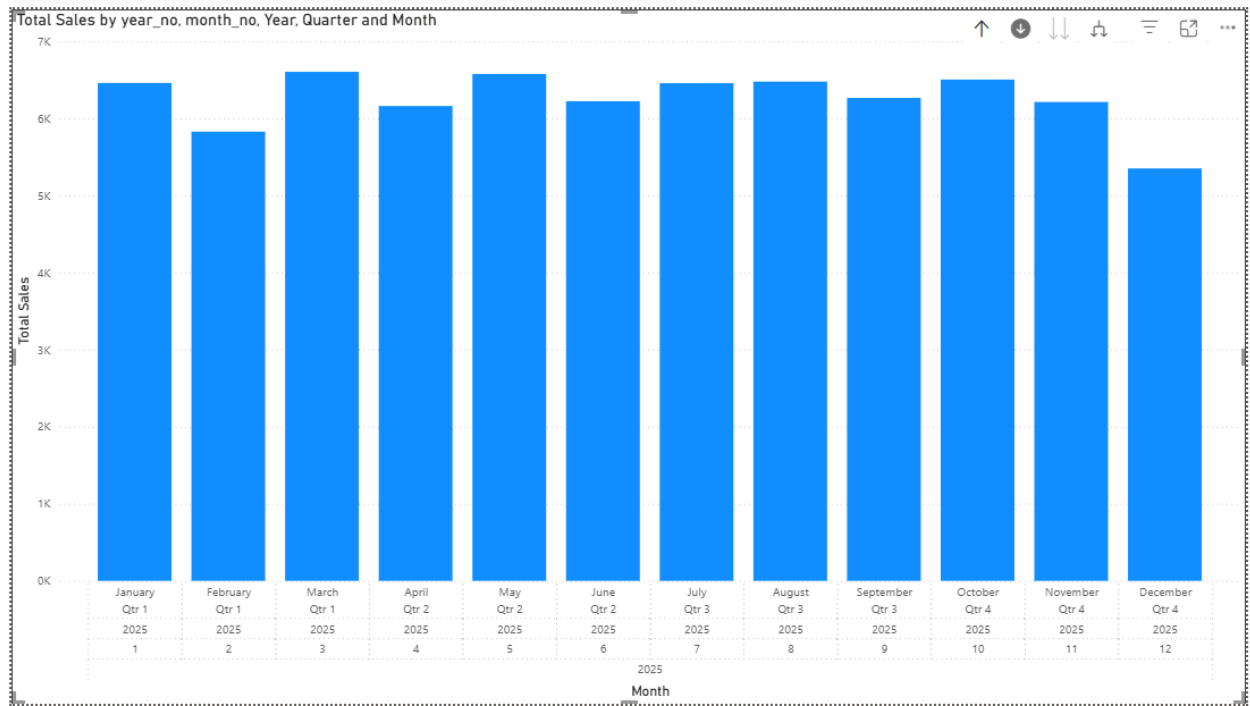
4.5. Report 3 — Drill-down report

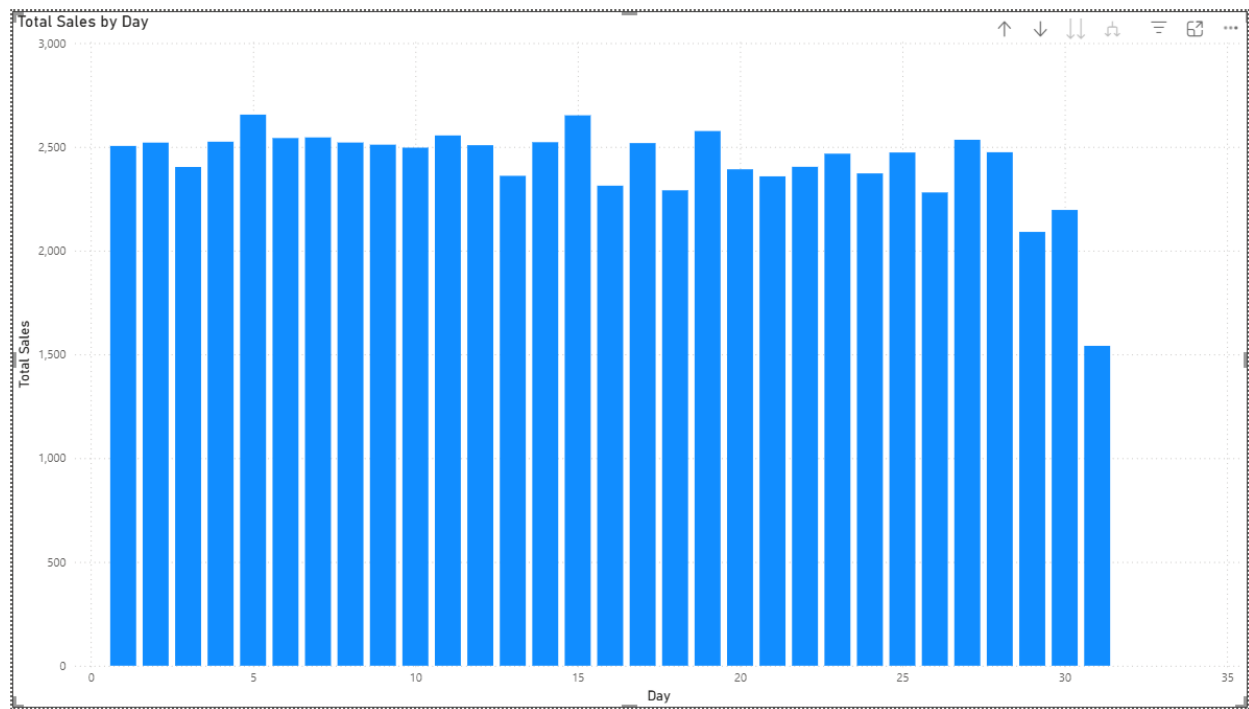
A drill-down report was created using a clustered column chart with a hierarchy of Year, Month, and Full Date. Total Sales was used as the measure, allowing users to drill from summary level to detailed level.

This report helps users move from yearly analysis into monthly and date-level detail to observe cinema sales trends more clearly.



Drill-down chart at the Year level.





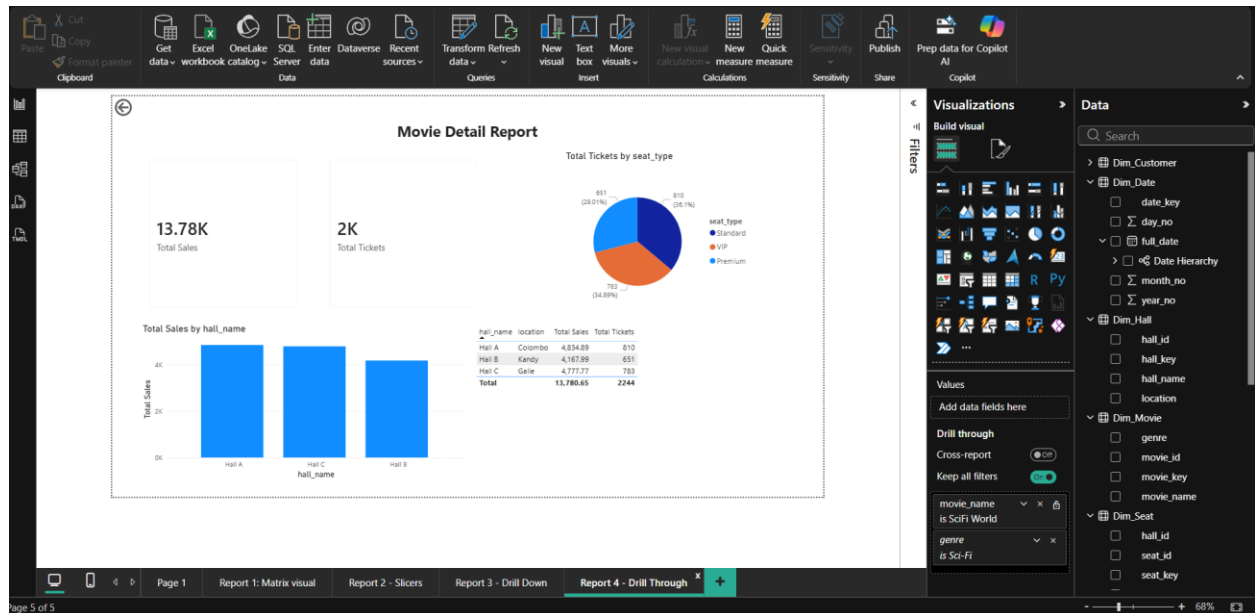
Drill-down chart expanded to Month or Full Date level.

4.6. Report 4- Drill-through report

A drill-through page was created using movie_name as the drill-through field. The page includes cards, charts, and a table to show detailed information for the selected movie. A back button was added to return to the previous report page.

When a user right-clicks a movie in another report and navigates to this page, all visuals are filtered to show detailed information only for the selected movie.

These four reports together demonstrate tabular analysis, filtering, hierarchical exploration, and detailed navigation using Power BI.



Drill-through detail page design.

Power BI Desktop interface showing the 'Report 1: Matrix visual' page. A context menu is open over the matrix, displaying options for drill-through. The 'Drill through' option is selected, leading to 'Report 4 - Drill Through'.

Matrix Data:

year_no	2025	Total Sales	Total Tickets	Total Sales	Total Tickets
genre					
Action		15,794.91	2604	15,794.91	2604
Comedy					
Drama					
Horror					
Sci-Fi					
Total					

Visualizations Panel:

- Build visual
- Filters
- Rows: genre, movie_name
- Columns: year_no, month_no
- Values: Total Sales, Total Tickets

Page Navigation: Page 1, Report 1: Matrix visual, Report 2 - Slicers, Report 3 - Drill Down, Report 4 - Drill Through

Power BI Desktop interface showing the 'Report 4 - Drill Through' page. The page displays a 'Movie Detail Report' with various visualizations and a data table.

Movie Detail Report Visuals:

- Total Sales:** 15.79K
- Total Tickets:** 3K
- Total Tickets by seat_type:** Pie chart showing distribution by seat type (Premium, VIP, Standard).
- Total Sales by hall_name:** Bar chart showing sales for Hall A, Hall B, and Hall C.

Data Table:

hall_name	location	Total Sales	Total Tickets
Hall A	Colombo	4,921.80	768
Hall B	Kandy	5,303.19	981
Hall C	Galle	5,569.92	855
Total		15,794.91	2604

Visualizations Panel:

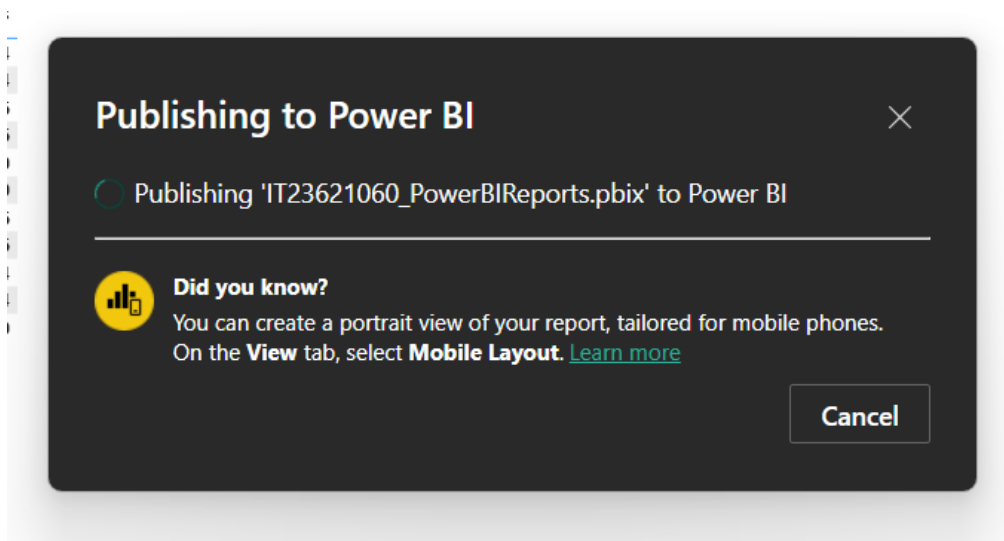
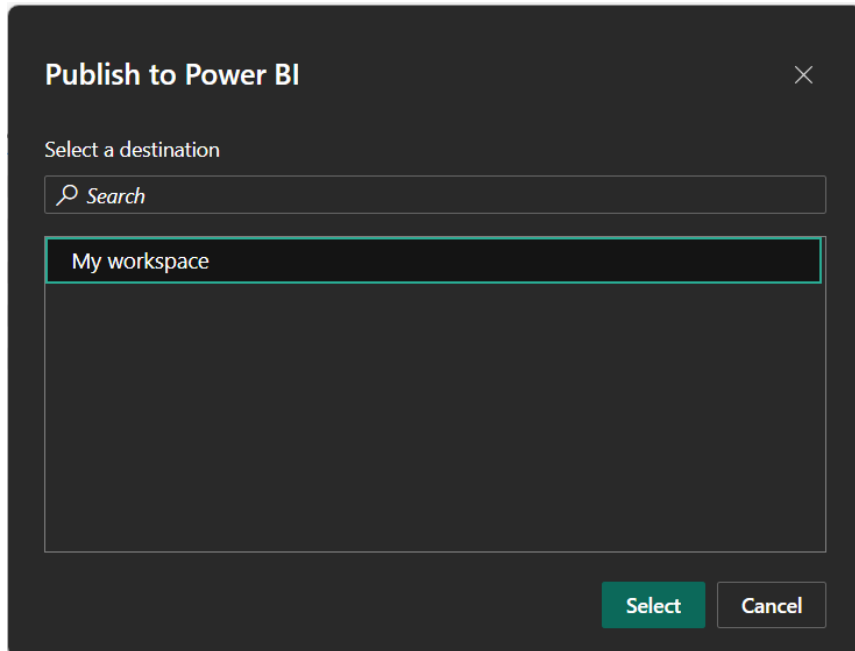
- Build visual
- Filters
- Values: Add data fields here
- Drill through: Cross-report (OFF), Keep all filters (ON)
- movie_name: is Action Blast
- genre: is Action

Page Navigation: Page 1, Report 1: Matrix visual, Report 2 - Slicers, Report 3 - Drill Down, Report 4 - Drill Through

Filtered drill-through page for the selected movie.

Step 5 - Publishing to Power BI Service

After completing the reports in Power BI Desktop, the .pbix file was published to Power BI Service using the Publish option. The report was uploaded to the online workspace and opened in the browser to confirm successful cloud deployment. This step satisfies the assignment requirement of demonstrating the reports in Power BI Service.



Publishing to Power BI



✓ Success!

[Open 'IT23621060_PowerBIReports.pbix' in Power BI](#)

[Get Quick Insights](#)



Did you know?

You can create a portrait view of your report, tailored for mobile phones.
On the **View** tab, select **Mobile Layout**. [Learn more](#)

Got it

Successful publishing of the Power BI report to Power BI Service.

Power BI

My workspace

Search

My workspace

+ New item

New folder

Import

Migrate

Filter by keyword

Filter

Workspace settings

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow

Add a task

Import a task flow

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity
IT23621060_PowerBIReports		Report		IT23621060 D...	4/2/2026, 11:22:5...			
IT23621060_PowerBIReports		Semantic mo...		IT23621060 D...	4/2/2026, 11:22:...	N/A		

IT23621060_PowerBIReports

Data updated 4/2/26

Search

Pages

File

Export

Share

Explore

Subscribe

Set alert

Monitor

Edit

Copilot

Page 1

Report 1: Matrix visual

Report 2 - Slicers

Report 3 - Drill Down

Report 4 - Drill Through

year_no

genre

2025

Total Sales

Total Tickets

Total

Total Sales

Total Tickets

Action

Action Blast

Comedy

Comedy Nights

Drama

Drama Life

Horror

Horror House

Sci-Fi

Sci-Fi World

Total

15,794.91

2604

15,794.91

2604

15,794.91

2604

14,880.96

2496

14,880.96

2496

14,880.96

2496

14,752.68

2520

14,752.68

2520

14,752.68

2520

15,889.56

2586

15,889.56

2586

15,889.56

2586

13,780.65

2244

13,780.65

2244

13,780.65

2244

75,098.76

12450

75,098.76

12450

Filters

IT23621060_PowerBIReports

Data updated 4/2/26

Search

Pages

File

Export

Share

Explore

Subscribe

Set alert

Monitor

Edit

Copilot

Page 1

Report 1: Matrix visual

Report 2 - Slicers

Report 3 - Drill Down

Report 4 - Drill Through

location

Colombo

Galle

Kandy

hall_name

Hall B

Total Tickets by genre

651 (16.48%)

981 (24.83%)

693 (17.54%)

840 (21.26%)

788 (19.89%)

genre

Action

Horror

Drama

Comedy

Sci-Fi

Total Sales by movie_name

6K

4K

2K

0K

Horror House

Action Blast

Drama Life

Comedy Nights

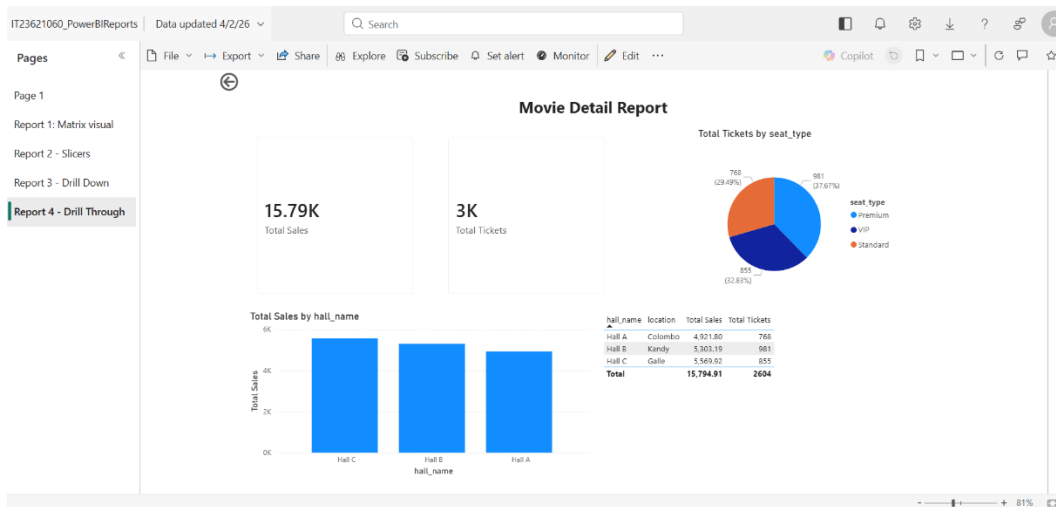
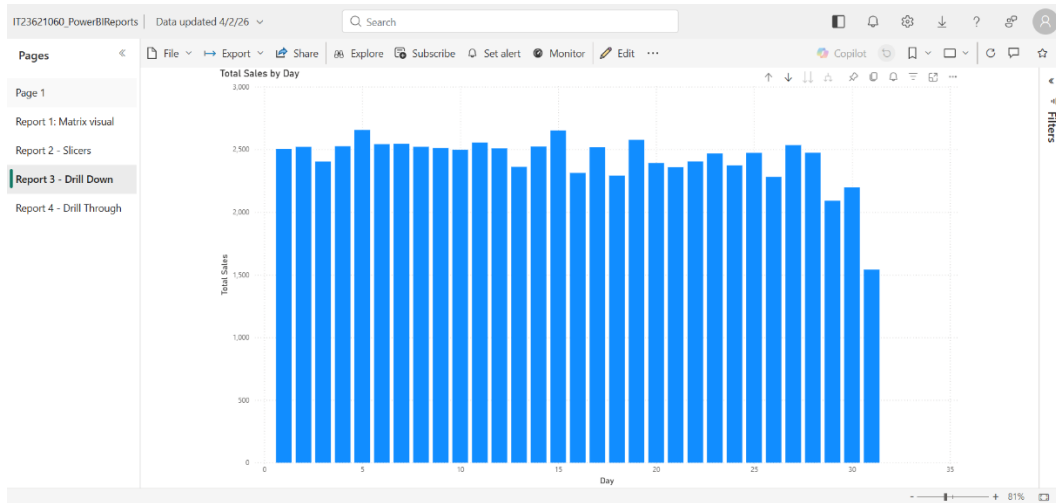
Sci-Fi World

movie_name

1K

Transaction Count

Filters



Published report viewed in Power BI Service (online).

Conclusion

The Assignment 1 data warehouse was successfully extended into a complete BI solution for Assignment 2. An SSAS cube was implemented with measures, dimensions, and a date hierarchy. OLAP operations were demonstrated in Excel using PivotTables, and interactive reports were created in Power BI Desktop and published to Power BI Service. The final solution supports meaningful analysis of cinema ticket sales and customer behavior.

References:

- Microsoft. *Install SQL Server Analysis Services*. Available at: <https://learn.microsoft.com/en-us/analysis-services/instances/install-windows/install-analysis-services?view=sql-analysis-services-2025>
- Microsoft. *Analysis Services documentation*. Available at: <https://learn.microsoft.com/en-us/analysis-services/>
- Microsoft. *Install SQL Server Data Tools (SSDT) for Visual Studio*. Available at: <https://learn.microsoft.com/en-us/sql/ssdt/download-sql-server-data-tools-ssdt?view=sql-server-ver17>
- Microsoft. *SQL Server Data Tools (SSDT)*. Available at: <https://learn.microsoft.com/en-us/sql/ssdt/sql-server-data-tools?view=sql-server-ver17>
- Microsoft. *Get Power BI Desktop*. Available at: <https://learn.microsoft.com/en-us/power-bi/fundamentals/desktop-get-the-desktop>
- Microsoft. *Get started with Power BI Desktop*. Available at: <https://learn.microsoft.com/en-us/power-bi/fundamentals/desktop-getting-started>
- Microsoft. *Data sources in Power BI Desktop*. Available at: <https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-data-sources>
- Microsoft. *Publish semantic models and reports from Power BI Desktop*. Available at: <https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-upload-desktop-files>
- Microsoft. *Power BI documentation*. Available at: <https://learn.microsoft.com/en-us/power-bi/>
- Kimball Group. *Dimensional Modeling Techniques*. Available at: <https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques/>